

EVERWELL MGTC0913E-20 SERVICE MANUAL





SERVICE MANUAL

Models

MGTC0913E-20 /MGTC0913C-20
MGTC0923E-20 /MGTC0923C-20
MGTC1213E-20 /MGTC1213C-20
MGTC1223E-20 /MGTC1223C-20
MGTC1823E-20 /MGTC1823C-20
MGTC2423E-20 /MGTC2423C-20
MGTC3623E-17 /MGTC3623C-17

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1. Important Notice

This service manual is intended for use by individuals possessing adequate backgrounds of electrical, electronic and mechanical experience. Any attempt to repair the appliance may result in personal injury and property damage. The manufacturer or seller cannot be responsible for the interpretation of this information, nor can it assume any liability in connection with its use.

The information, specifications and parameter are subject to change due to technical modification or improvement without any prior notice. The accurate specifications are presented on the nameplate label.

How to order spare parts

To have your order filled promptly and correctly, please furnish the following information:

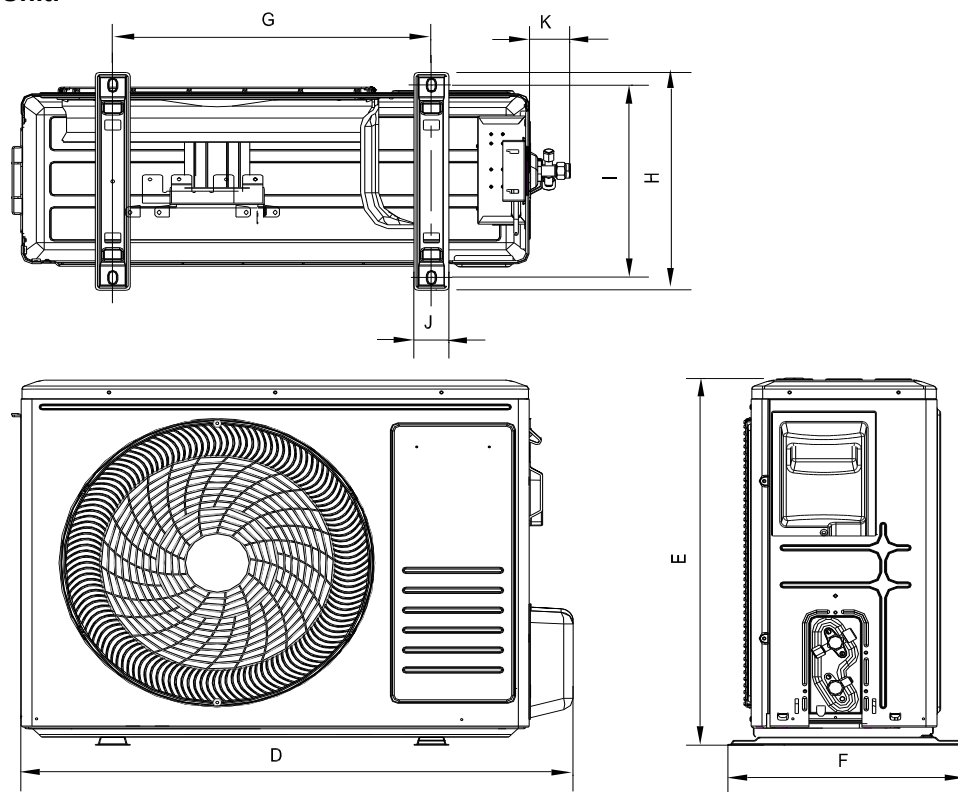
1. Model No. with Indoor or Outdoor
2. No. in the Explosion View
3. Part Name
4. The quantity you ordered

2. Product Dimensions

Indoor Unit:



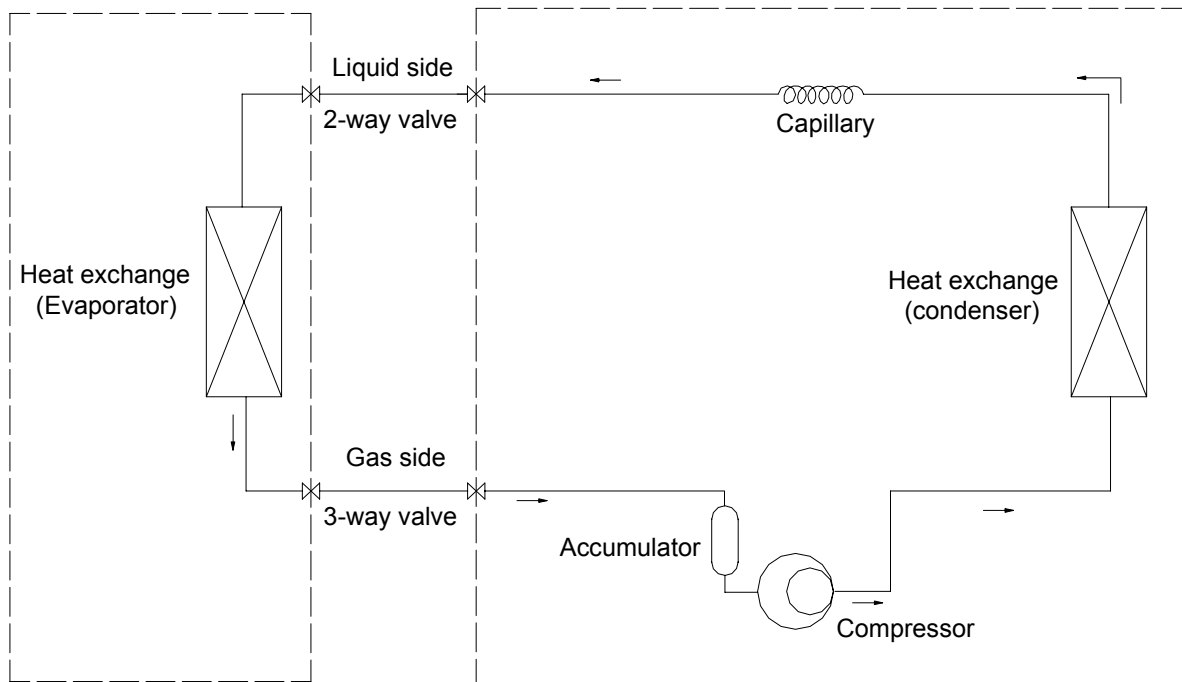
Outdoor Unit:



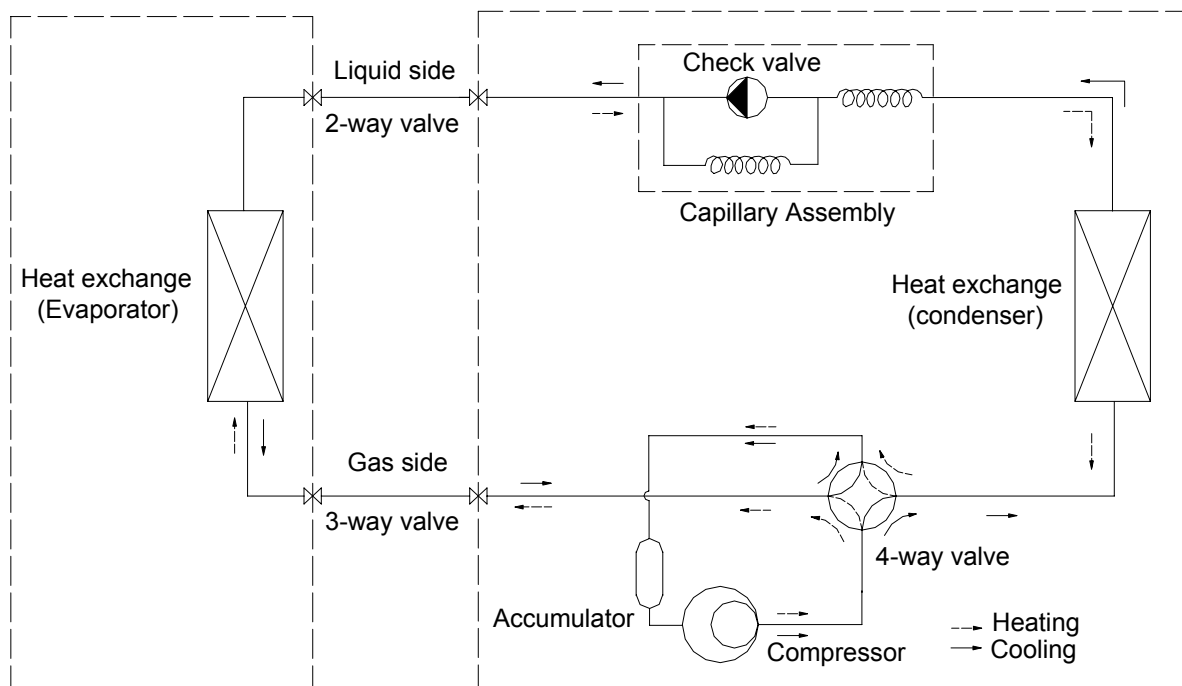
Model	Indoor unit			Outdoor unit							
	A	B	C	D	E	F	G	H	I	J	K
MGTC0913E-20 /MGTC0913C-20	811	203	292	810	549	305	433.8	305	278	48	55
MGTC0923E-20 /MGTC0923C-20	811	203	292	787	498	290	415	291	225	48.5	52
MGTC1213E-20 /MGTC1213C-20	811	203	292	810	549	305	433.8	305	278	48	55
MGTC1223E-20 /MGTC1223C-20	811	203	292	787	498	290	415	291	225	48.5	52
MGTC1823E-20 /MGTC1823C-20	1010	220	315	863	602	349	518	357	329	55	63
MGTC2423E-20 /MGTC2423C-20	1186	258	340	927	699	380	586	374.9	347.5	58.6	63
MGTC3623E-17 /MGTC3623C-17	1186	258	340	975	808	433	675	399	378	50	63

3. Refrigeration cycle diagram

Cooling only

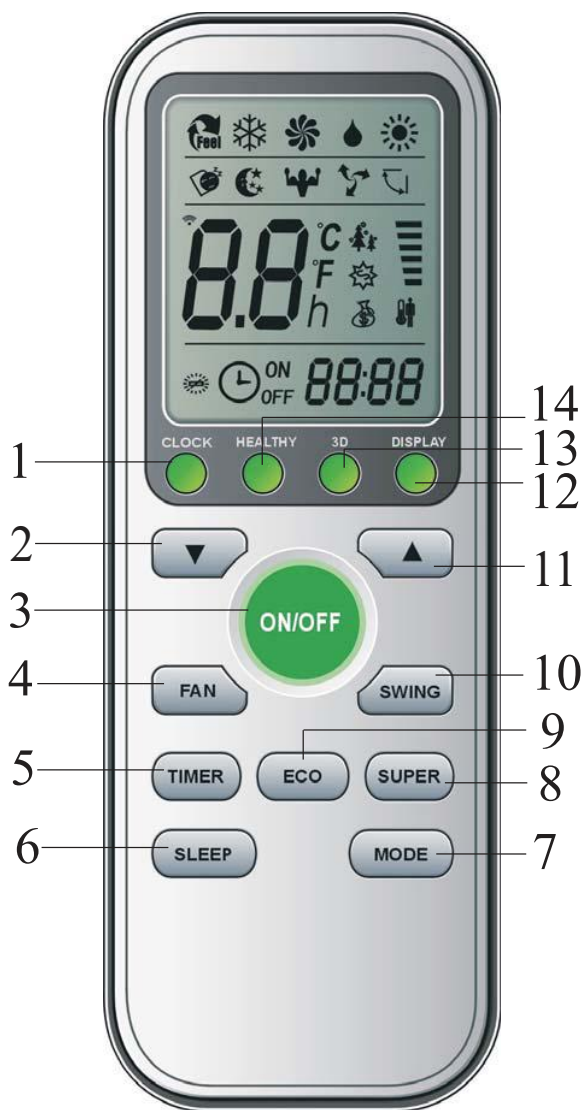


Heat pump



4.Operation Details

Remote controller



- 1 CLOCK**
Press this button to set the time. When the time is flashing, press ▲ or ▼ to adjust the time. Every press equals one minute, or press and hold for rapid adjustment. Press CLOCK again once the time has been set.
- 2 TEMP DOWN button**
Decrease the temperature or time by 1 unit.
- 3 ON/OFF button**
To switch the conditioner on and off.
- 4 FAN SPEED button**
To select the fan speed of auto/low/mid/high.
- 5 TIMER button**
To set automatic switching-on/off.
- 6 SLEEP button**
To activate the function "SLEEP".
- 7 MODE button**
To select the mode of operation.
- 8 SUPER button**
In cooling mode, press this button, the unit will give the maximum cooling temperature with 16°C.
In heating mode, press this button, the unit will give the maximum heating temperature with 31°C.
- 9 ECO button**
In cooling mode, press this button, the temperature will increase 2°C on the base of setting temperature.
In heating mode, press this button, the temperature will decrease 2°C on the base of setting temperature.
- 10 SWING button**
To activate or deactivate of the movement of the "DEFLECTORS".
- 11 TEMP UP button**
Increase the temperature or time by 1 unit.
- 12 DISPLAY button**
To switch on/off the LED display (if present).
- 13 3D**
When you press "3D", the horizontal and vertical vanes will swing together at the same time.
- 14 HEALTHY button**
To switch - on /off HEALTHY function. It is a button which controls the ionizer or plasma generator only for inverter type.

Note: Each mode and relevant function will be further specified in following pages.

Remote Control

The remote controller is not presetting as Cooling Only Air Conditioner or Heat Pump by manufacturer. Each time after the remote controller replace batteries or is energized, the arrowhead will flashes on the front of "Heat" or "Cool" on LCD of the remote controller.

User can preset the remote controller type depending on the air conditioner type you have purchased as follows: Press any button when the arrowhead flashes on the front of "Cool", Cooling Only is set. Press any button when the arrowhead flashes on the front of "Heat", Heat Pump is set.

If you don't press any button within 10 seconds, the remote controller is preset as Heat Pump automatically.

Note :

If the air conditioner you purchased is a Cooling Only one, but you preset the remote controller as Heat Pump, it doesn't bring any matter. But if the air conditioner you purchased is a Heat Pump one, and you preset the remote controller as Cooling Only, then you CAN NOT preset the Heating operation with the remote controller.

Electronic Controller

1. Safety Protection

- (1) Time Delay for Safety protection
 - 3 minutes delay for compressor ---The compressor is ceased for 3minutes before restarting to balance the pressure in the refrigeration cycle in order to protect the compressor.
 - 2 minutes delay for 4-way valve---The 4-way valve will be ceased for 2 minutes late after compressor to prevent the refrigerant-gas abnormal noise when the HEATING operation is OFF or switch to the other operation mode.
- (2) Discharge temperature protection
There is a temperature sensor on top of compressor, when temperature on top of compressor exceeded the limit, system control will shut down the compressor and the display board will show the error code.
- (3) Lower voltage protection
When AC voltage $\leq 158V$ and keep it for 10 seconds, unit will be shut down for protection.
- (4) Over voltage protection
When AC voltage $\geq 260V$, unit will be shut down and recover while $AC \leq 255V$.
- (5) Over current protection
When the current of outdoor unit is overload, controller shut down the unit immediately and show error code.
- (6) Compressor abnormality protection
When compressor start on or in the process of running, if there is no feedback to controller or load of compressor is abnormality, the air conditioner will shut down, and show error code.
- (7) **IPM** module protection
IPM module has high temperature & over current protection itself, if there is signal feedback to IPM, the outdoor unit will shut down, LED on outdoor PCB will show the error code.

2. “Feel” Mode Operation

- (1) When the “Feel” mode is selected, the operation mode and initial temperature set are determined by the initial room temperature at start-up of the operation except to turn off the air conditioner and operates it again.
- (2) If the mode is change to “Feel” from other mode, the “Feel” mode doesn’t operate until compressor stop for more than 3 minutes.

Mode	Initial Room Temperature	Initial Set Temperature
COOLING	$RT = 26^{\circ}C$	$23^{\circ}C$
DRY	$26^{\circ}C > RT \geq 20^{\circ}C$	$RT - 2^{\circ}C$
HEATING for Heat Pump/ FAN for Cooling Only	$RT < 20^{\circ}C$	-

- In the “Feel” mode, when the controller receives the up or down signal of temperature, the set temperature can adjust by $1^{\circ}C$ upper or lower. The biggest you can adjust by $2^{\circ}C$ upper or lower.

3. “COOLING” Mode Operation

- (1) Compressor frequency control
According to difference room temperature and set temperature ($\delta t = RT - ST$), running frequency of compressor is controlled by electronic controller. When room temperature is much higher than set temperature, the compressor will start at a high frequency, and as room temperature goes down, the compressor running frequency will go down. When room temperature is lower than set temperature,

the compressor will run at very low frequency. In general, unit will change its running frequency according to δt to make room temperature closing to set temperature.

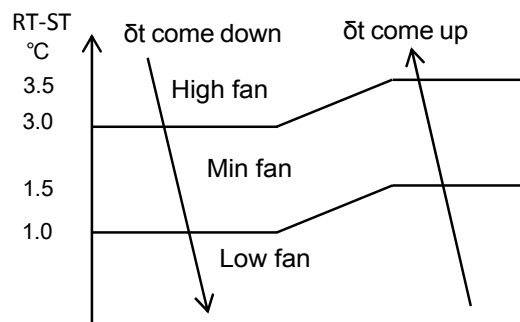
(2) Outdoor temperature affects running frequency of the compressor

Outdoor temperature affect compressor's running frequency. Difference inlet temperature of outdoor unit is adapted by difference compressor running frequency. While outdoor temperature is about 30°C, the compressor will run in high frequency.

If unit run in "cooling" mode and outdoor temperature is less than -2°C, the controller will shut down compressor and show error code, while the ambient temperature is over 1°C, the compressor will run automatically.

(3) Auto fan control in cooling mode

In cooling mode (include cooling in "feel" mode), fan speed is determined by δt , as the following diagram:



4. "DRY" Mode Operation

(1) The system for DRY operation used the same refrigerant circle as the cooling one.

(2) When the system operates in DRY mode, at first it operates in cooling mode, the set temperature (ST) is "RT-2°C". After that, the system will operate in cooling mode with lowest fan speed for 30 minutes and then change to middle speed for 30seconds as a cycle. During the course of this operation, you can't use remote controller to adjust the fan speed but you can control the vane direction.

(3) In the dry mode, when $RT \leq 12^\circ\text{C}$, the compressor will stop and operates again at $RT \geq 14^\circ\text{C}$.

5. "HEATING" Mode Operation (available for Heat Pump only)

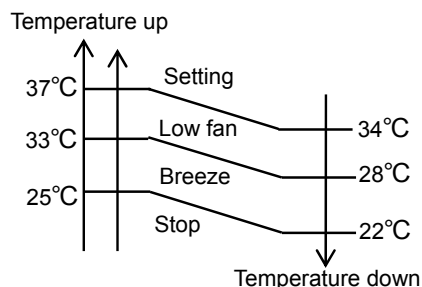
(1) Frequency control

Same as the frequency control in cooling mode, the running frequency of compressor is controlled by controller. Unit change its running frequency according to δt to make the room temperature closing to the set temperature.

(2) Indoor fan motor control

1) Cold Air Prevention Control

- The function is intended to prevent cold air from being discharged when heating mode selected or while in defrosting.
- The indoor fan speed will be controlled as following.

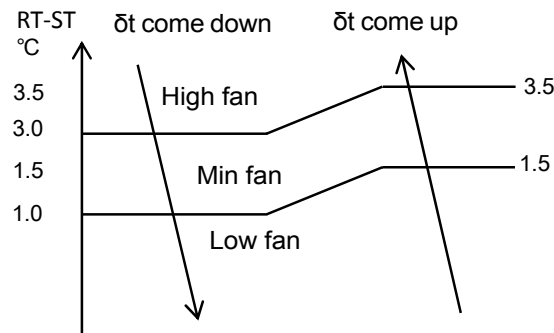


- In heating operation, if air conditioner turn off, the indoor fan motor will run most for 30

seconds since the stop of compressor.

2) Auto fan control (heating)

In heating mode (include in "I feel" mode), fan speed is determined by δt as the following:



(3) 4-way valve control

In heating mode, 4-way valve will power on ahead of compressor for 5 seconds, and cut off for 2 minutes later than compressor. 4-way valve will not power off unless the machine to be switched off, mode changed or on the process of defrosting.

(4) Defrosting

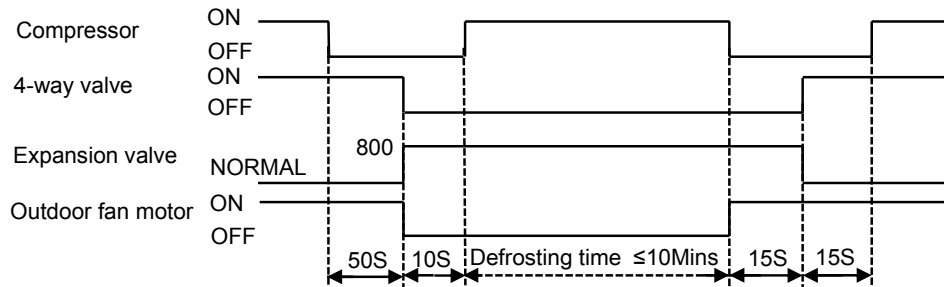
Defrosting is controlled by the microprocessor.

When one of the following conditions is satisfied, unit comes into defrosting:

- Outdoor heat exchanger Temperature (OPT) is continuously less than $3^{\circ}C$ while the unit runs for more than 40 minutes, and OPT is keeping under $-6^{\circ}C$ for more than 3 minutes.
- Outdoor heat exchanger Temperature (OPT) is continuously less than $3^{\circ}C$ meanwhile the unit runs for more than 80 minutes, and OPT is keeping under $-4^{\circ}C$ for more than 3 minutes.
- Outdoor heat exchanger Temperature (OPT) is continuously less than $3^{\circ}C$ while the unit runs for more than 120 minutes, and OPT is keeping below $-2^{\circ}C$ for more than 3 minutes.

Before the air con comes into defrosting, compressor running frequency drop down to a lower frequency firstly, then the compressor shuts down.

In defrosting, the max. frequency of compressor is F9 (a little less than the highest frequency). In this period all protection function are available.



In defrosting, LED showing by winking.

No matter what AC come into or out of defrosting, indoor fan motor speed is the same as Cold Air Prevention Control.

While one of the following conditions is satisfied, unit comes out of defrosting and shifts to heating mode:

- Outdoor coil Temperature (OPT) $\geq 15^{\circ}C$.
- The time of defrosting keeps for more than 10 minutes.

(5) Indoor exchanger overheat protection

When Indoor exchanger Temperature (IPT) is higher than $55^{\circ}C$, unit comes into indoor exchanger

overheat protection. The compressor drops its frequency toward to F1 level until $IPT \leq 52^{\circ}\text{C}$

If $IPT \leq 52^{\circ}\text{C}$ and keep for 5 minutes, control system doesn't limit running frequency.

If $IPT > 62^{\circ}\text{C}$, control system shut down compressor, and recover while IPT drop less than 50°C .

6. "SLEEP" mode

When the SLEEP button is pressed, the AC operates as following:

- The indoor fan speed is set at low speed, the power lamp and the sleep lamp is on, the display of temperature will close after 5 minutes.
- When selecting COOLING/DRY operation with SLEEP mode, the set temperature will be raised by 1°C 1 hour later and by 2°C 2 hour later.
- When selecting HEATING operation with SLEEP mode, the set temperature will be dropped by 1°C 1 hour later and 2°C 2 hours later.
- After the System operates in SLEEP mode for 8 hours, it will stop automatically.

7. EMERGENCY Operation

When the EMERGENCY Operation switch is pressed one time, COOLING mode is selected and if the EMERGENCY Operation switch press again within 3s, HEATING mode selected, while press once again, the unit will switch off.

When the remote controller missing, failed or the batteries run down, press the EMERGENCY Operation switch on front of the indoor unit for function test.

NOTE: Do not press the EMERGENCY Operation switch during normal operation.

8. AUTO-RESTART Function (Option)

While air conditioner is operating in one mode, all of its operation data, such as working mode, preset temperature etc. would be memorized into IC by main PCB. If power supply cut off due to reasons and recover again, the AUTO-RESTART function will set synchronously and the air conditioner would work at the same mode as before.

Auto-restart Pre-setting (optional):

If Auto-restart function is needed, follow the steps below to activate this function:

- 1) Pulling the air-con's plug out of socket.
- 2) Pressing and holding the Emergency button (ON/OFF) on the indoor, then insert the plug into the socket again.
- 3) Keep pressing the Emergency button for more than 10 seconds until three short beeps heard, the Auto-restart function been activated.

9. Protection and Failure Display

- When protection display is available, controller will show error code, digital LED shows error code and setting temperature by turns.
- If there is more than one failure, it will show error codes according to the error list sequence.
- To insure the signal communication of indoor and outdoor unit, any failure code relates to outdoor unit will remain display for 2 minutes maximum after it's recovered.
- Among all the failure codes, sensor failure can be recovery automatically once it comes normal.

● Error list

Failure Type	LED Code	Digital LED Code
Indoor and outdoor communication failure	RUN、TIMER—both winking	E0
Outdoor communication failure	RUN、TIMER—both winking	EC
Indoor room temperature sensor (IRT)	RUN-1 time/8s	E1
Indoor coil temperature sensor (IPT)	RUN-2 times /8s	E2
Outdoor coil temperature sensor (OPT)	RUN-3 times /8s	E3
System abnormality	RUN-4 times /8s	E4
Model configuration wrong	RUN-5 times /8s	E5
Indoor fan motor fault	RUN-6 times /8s	E6
Outdoor temperature sensor	RUN-7 times /8s	E7
Exhaust temp. sensor	RUN-8 times /8s	E8
IPM drive and module fault	RUN-9 times /8s	E9
Outdoor fan motor fault (DC motor)	RUN-10 times /8s	EF
Current sensor fault	RUN—11 times /8s	EA
Indoor unit EEPROM fault	RUN-12 times /8s	EE
Temp. switch fault (on top of the compressor)	RUN-13 times /8s	EP
Voltage sensor fault	RUN-14 times /8s	EU

● Protection display code list:

Protection Type	Function Indicator (flash)	Digital LED Code
Overvoltage /lower voltage protection	RUN: Blink; TIMER: 1 blink /8 sec	P1
Overcurrent protection	RUN: Blink; TIMER: 2 blink /8 sec	P2
Exhaust over temperature protection	RUN: Blink; TIMER: 4 blink /8 sec	P4
Too cool protection in cooling mode	RUN: Bright; TIMER: 5 blink /8 sec	P5
Overheat protection in cooling mode	RUN: Bright; TIMER: 6 blink /8 sec	P6
Overheat protection in heating mode	RUN: Bright; TIMER: 7 blink /8 sec	P7
Outdoor over temperature / lower temperature protection	RUN: Bright; TIMER: 8 blink /8 sec	P8
Drive protection (software control)	RUN: Blink; TIMER: 9 blink /8 sec	P9
Module protection (hardware control)	RUN: Blink; TIMER: 10 blink /8 sec	P0

● Outdoor failure display

There is a LED on outdoor power board, it blinks 1s ON and 1s OFF while compressor standby and it always light (ON) while compressor running; If there is failure happened on ODU, The indicator (LED) alerts the fault in a cycle as such that it is bright for 0.5 seconds, dark for 0.5 seconds, blinks “n” times and then dark for 3 seconds. For details as table below:

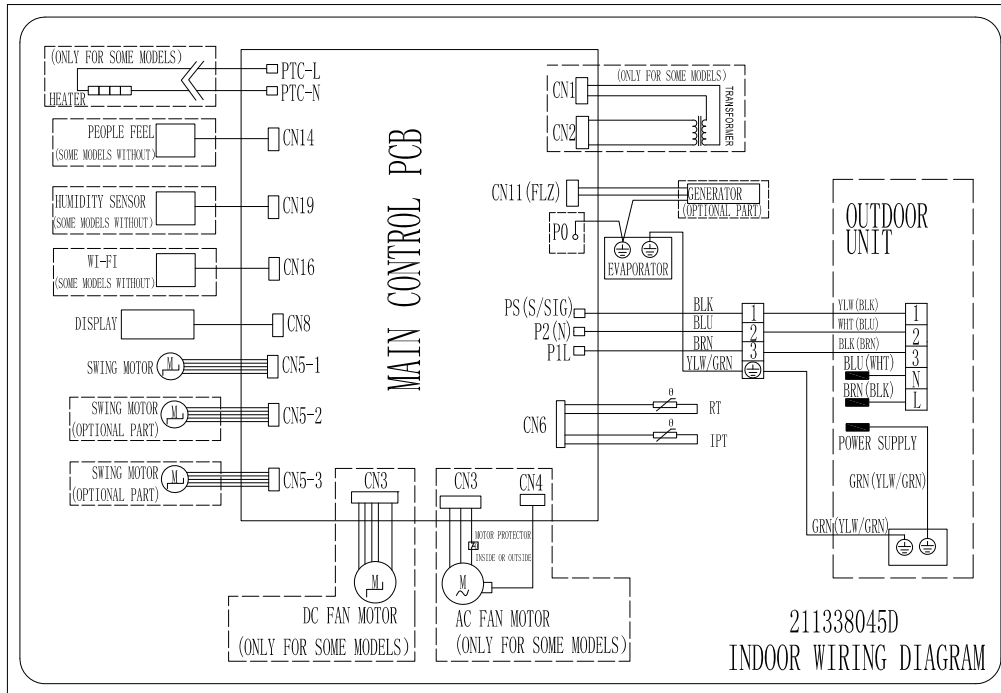
Display on outdoor power source board:

blink time (n)	Failure
1	IPM protection
2	Over voltage /lower voltage
3	Overcurrent
4	Exhaust over temperature protection
5	Outdoor coil over temperature protection
6	Drive fault and protection (V1, VP1)
7	Communication fault with indoor unit
8	Compressor overheat fault (compressor top switch)
9	Short-circuit / open-circuit fault of outdoor temperature sensor
10	Short circuit / open-circuit fault of outdoor heat exchanger temperature sensor
11	Short-circuit / open-circuit fault of exhaust temperature sensor
12	Voltage sensor fault
13	Current sensor fault
14	IPM fault
15	Communication fault between power source board and IPM
16	No feedback from DC fan motor(outdoor unit)
17	Defrost state

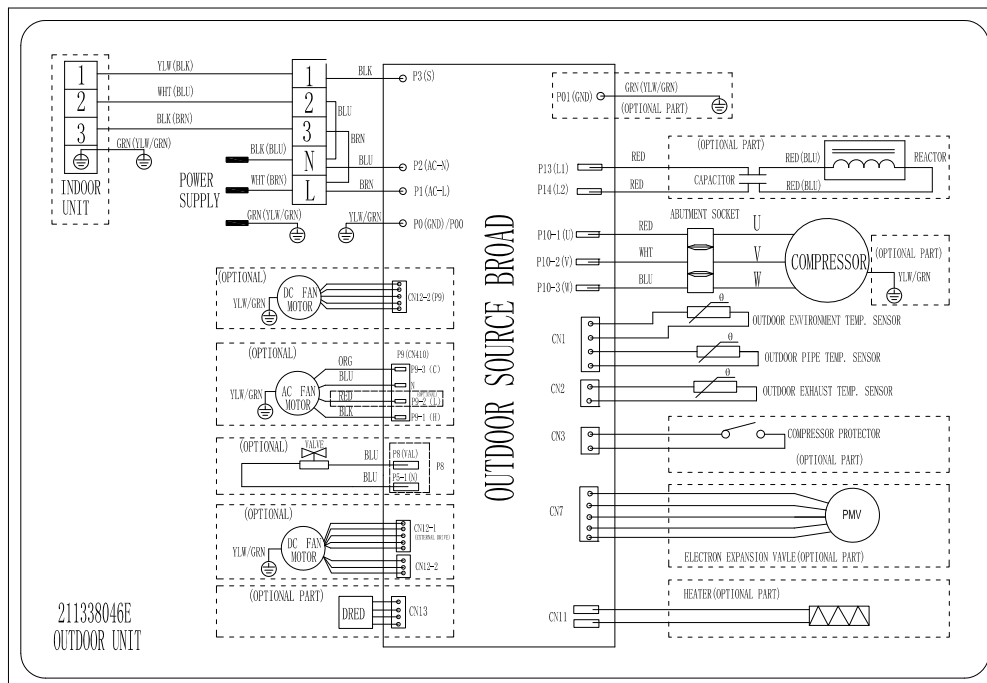
5. Wiring diagram

Mode: MGTC0913E-20 /MGTC0913C-20

INDOOR UNIT:

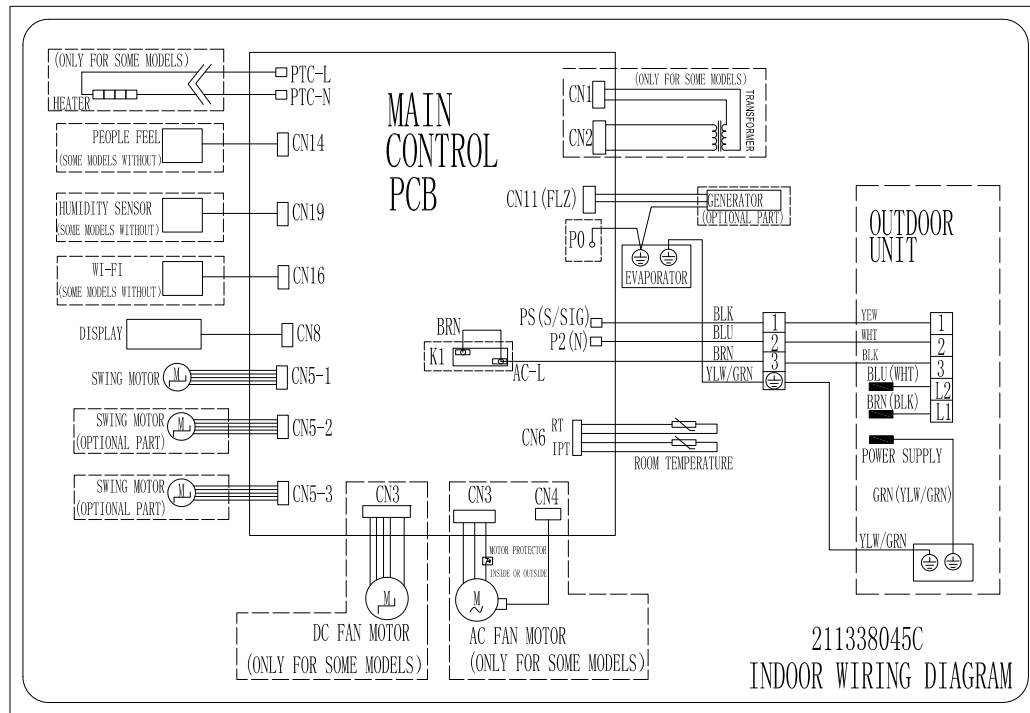


OUTDOOR UNIT

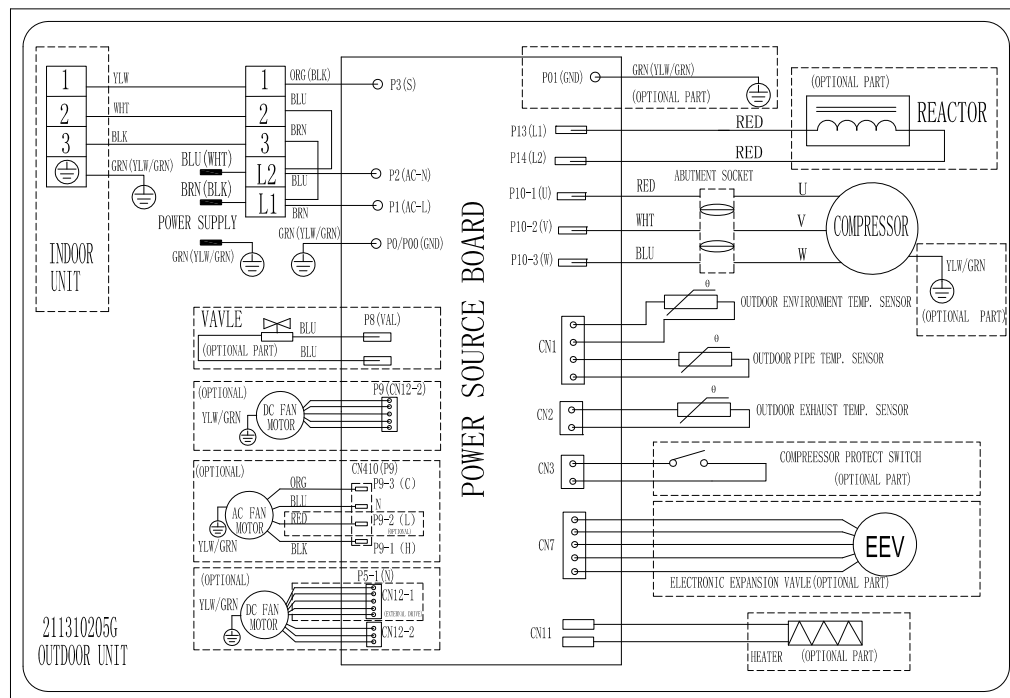


Mode:MGTC0923E-20 /MGTC0923C-20

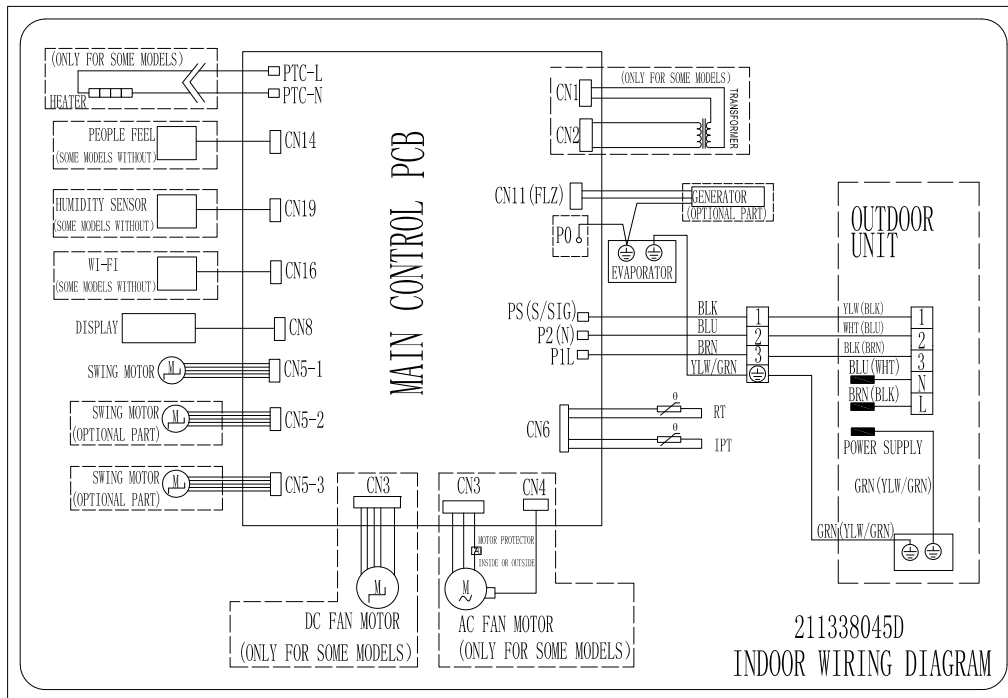
INDOOR UNIT



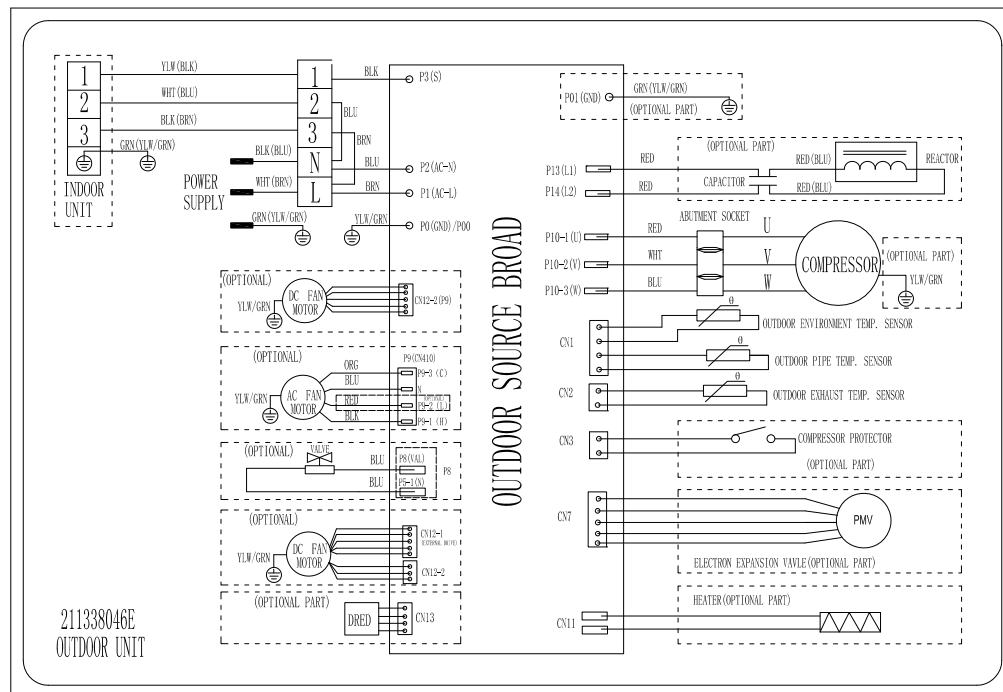
OUTDOOR UNIT



Mode: MGTC1213E-20 /MGTC1213C-20
INDOOR UNIT

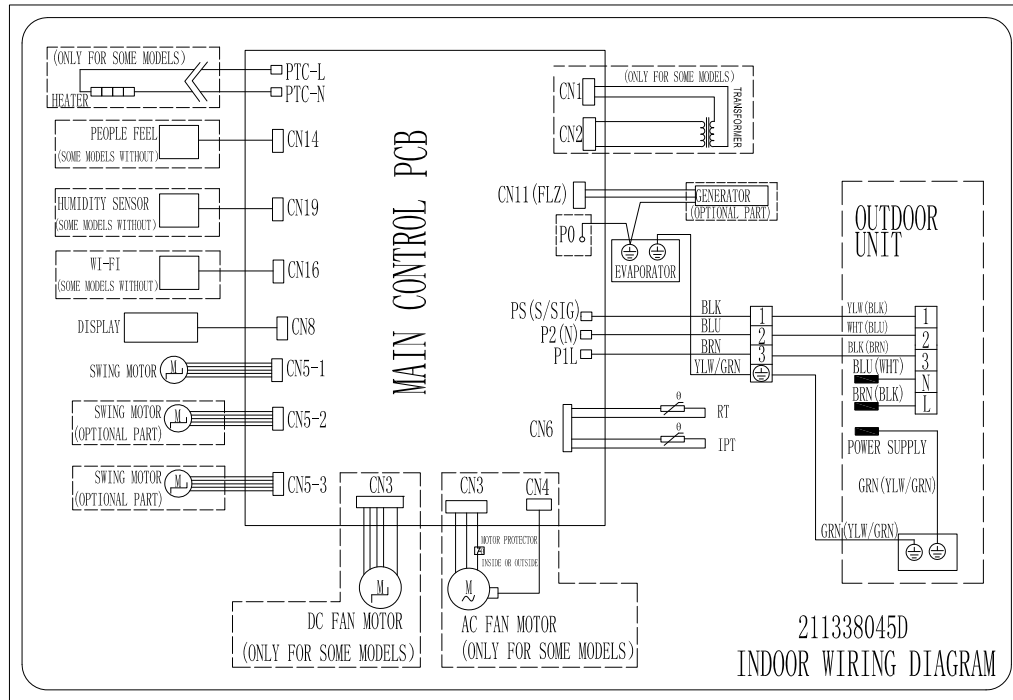


OUTDOOR UNIT

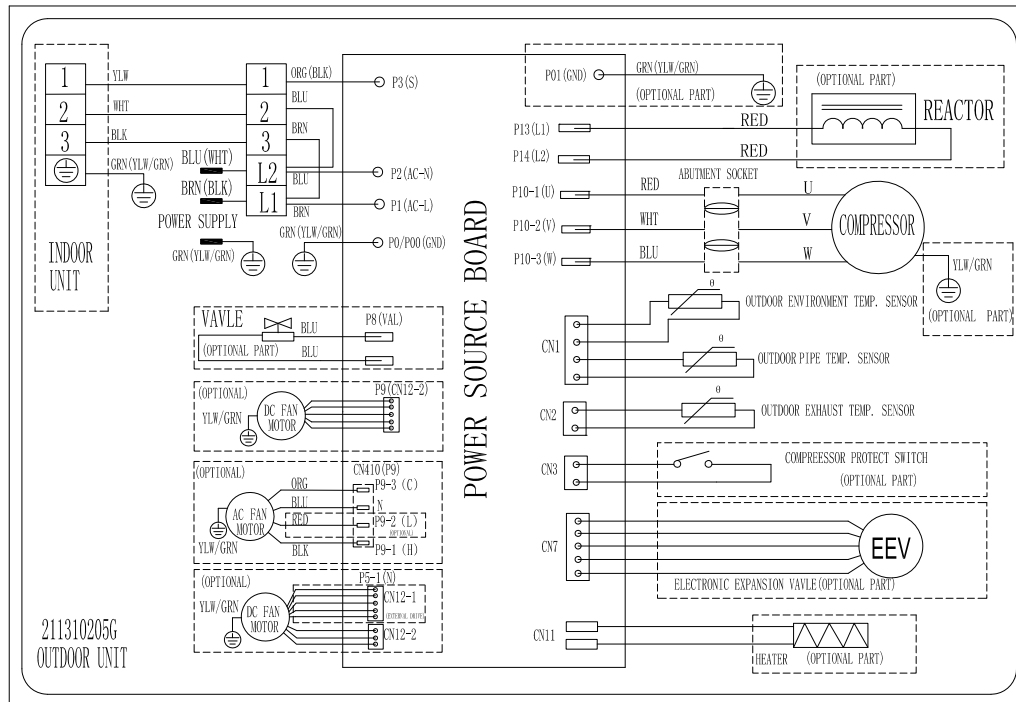


MODEL: MGTC1223E-20 /MGTC1223C-20

INDOOR UNIT

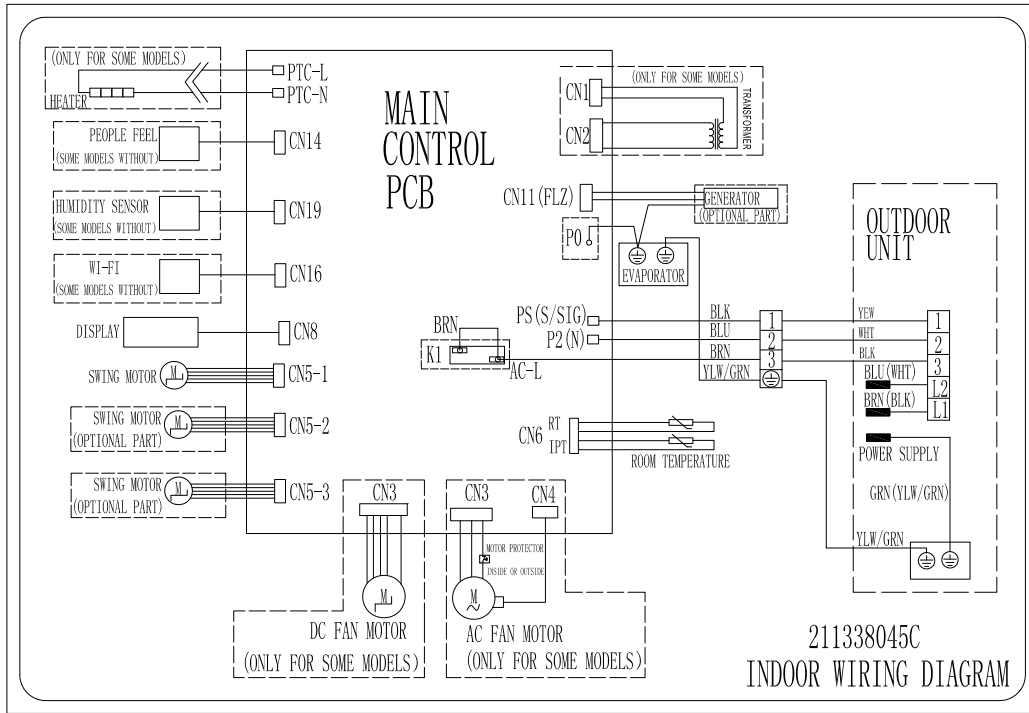


OUTDOOR UNIT

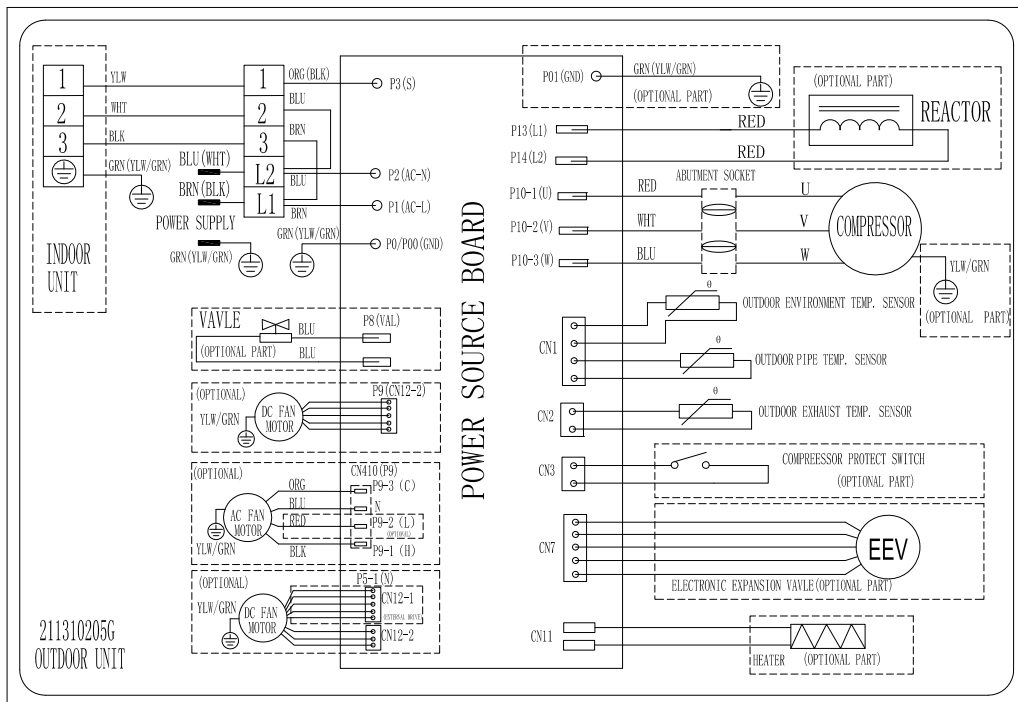


MODEL: MGTC1823E-20 /MGTC1823C-20, MGTC2423E-20 /MGTC2423C-20,
MGTC3623E-17 /MGTC3623C-17

INDOOR UNIT

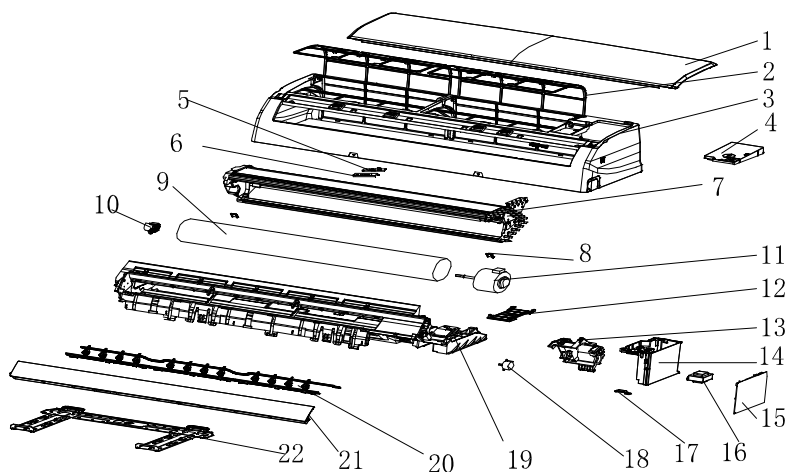


OUTDOOR UNIT

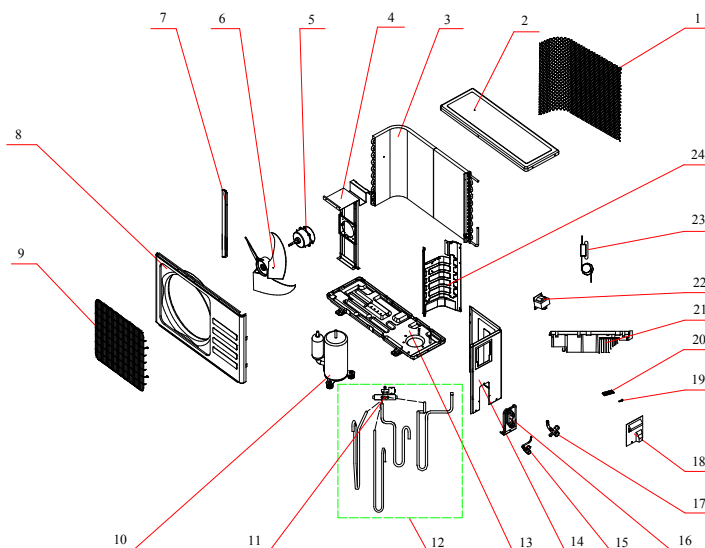


6. Explosion view and parts

Indoor Unit- G4BMV00000 KFR-26G/YXABp(E1/6)(BMV00)

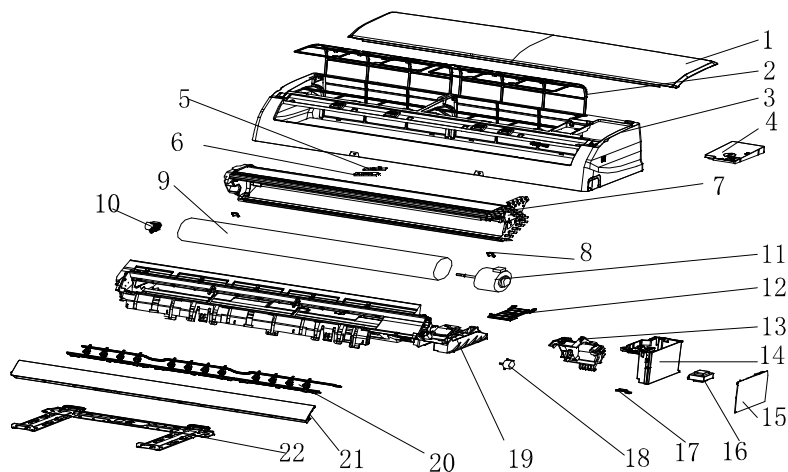


No.	Part No.	Part Name	Q'ty	Remark
1		Front Panel	1	
2		Air Filter	2	
3		Face Frame	1	
4		Electrical Box Cover	1	
5		Display PCB	1	
6		Display PCB Box	1	
7		Evaporator	1	
8		Screw Cover	2	
9		Cross Fan	1	
10		Bearing Mount	1	
11		Indoor Motor	1	
12		In And Out Pipe Fixer	1	
13		Indoor Motor Cover	1	
14		Electrical Box	1	
15		Main PCB	1	
16		Transformer	1	
17		Cable Clamp	1	
18		Vane Motor 1	1	
		Vane Motor 2	1	
19		Base	1	
20		Vertical Vane Assembly	2	
21		Vane	1	
22		Installation Plate	1	
23		Indoor Sensor Assembly	1	Not shown in Explosion view
24		Remote Controller	1	
25		wifi	1	
26		Drainage Hose	1	
27		Indoor Carton	1	
28		Left Foaming	1	
29		Right Foaming	1	

Outdoor Unit- G5BBW00000 KFR-26W/YXABp(E1/6)(BBW00)

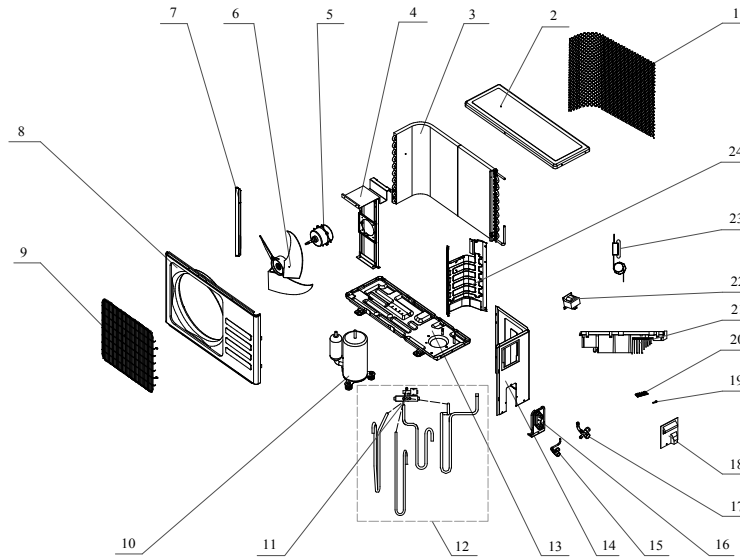
No.	Part No.	Part Name	Q'ty	Remark
1		Grille	1	Optional
2		Top Cover	1	
3		Condenser	1	
4		Outdoor Motor Supporter	1	
5		Outdoor Motor	1	
6		Propeller Fan	1	
7		Left grille supporter	1	
8		Front Plate	1	
9		Fan Guard	1	
10		Compressor	1	
11		4-way Valve	1	
12		4-way Valve Assembly	1	
13		Base	1	
14		Right Plate	1	
15		Two-way Valve	1	
16		Valve Supporter	1	
17		Three-way Valve	1	
18		Electrical Box Cover	1	
19		Terminal	1	
20		Cable clamp1	1	
21		Outdoor PCB Assembly	1	
22		Inductor	1	
23		Capillary Assembly	1	
24		Partition plate	1	
25		Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26		Discharge Temp. sensor	1	
27		Compressor Capacitor	1	
28		Capacitor Strip	1	
29		Base carton	1	
30		Cabinet carton	1	
31		Base foaming	1	
32		Cover foaming	1	

Indoor Unit- MGTC0923E-20



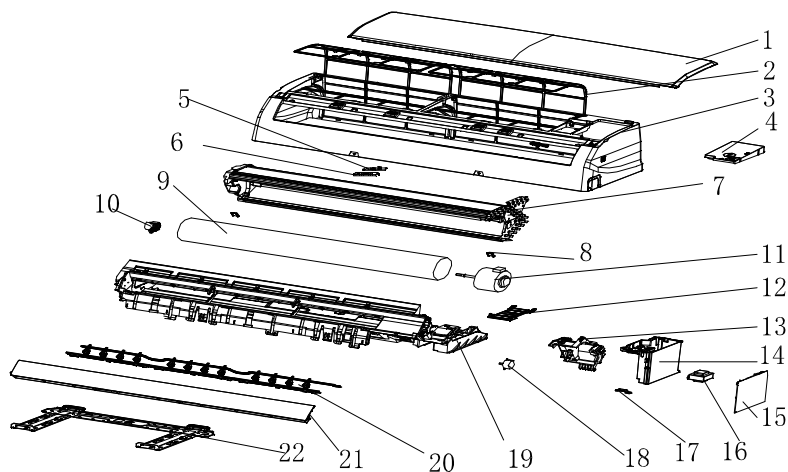
No.	Part No.	Part Name	Q'ty	Remark
1	41106-003591	Front Panel	1	
2	42008-000039	Air Filter	2	
3	41106-002375	Face Frame	1	
4	41211-000077	Electrical Box Cover	1	
5	31102-000084	Display PCB	1	
6	42003-000001	Display PCB Box	1	
7	92011-004784	Evaporator	1	
8	41108-000066	Screw Cover	2	
9	42004-000001	Cross Fan	1	
10	42007-000001	Bearing Mount	1	
11	22001-000267_UM	Indoor Motor	1	
12	41101-000242	In And Out Pipe Fixer	1	
13	42003-000051	Indoor Motor Cover	1	
14	41105-000134	Electrical Box	1	
15	31101-000247_UM	Main PCB	1	
16	NO	Transformer	1	
17	42001-000103	Cable Clamp	1	
18	22001-000313	Vane Motor 1	1	
19	41102-000091	Base	1	
20	41101-000082	Vertical Vane Assembly	2	
21	41103-000102	Vane	1	
22	41109-000041	Installation Plate	1	
23	10104-100014	Indoor Sensor Assembly	1	Not shown in Explosion view
24	22013-000946	Remote Controller	1	
25	32001-000057	Wifi	1	
26	A1101-007281	Indoor Carton	1	
27	41110-000217	Left Foaming	1	
28	41110-000262	Right Foaming	1	

Outdoor Unit- MGTC0923C-20



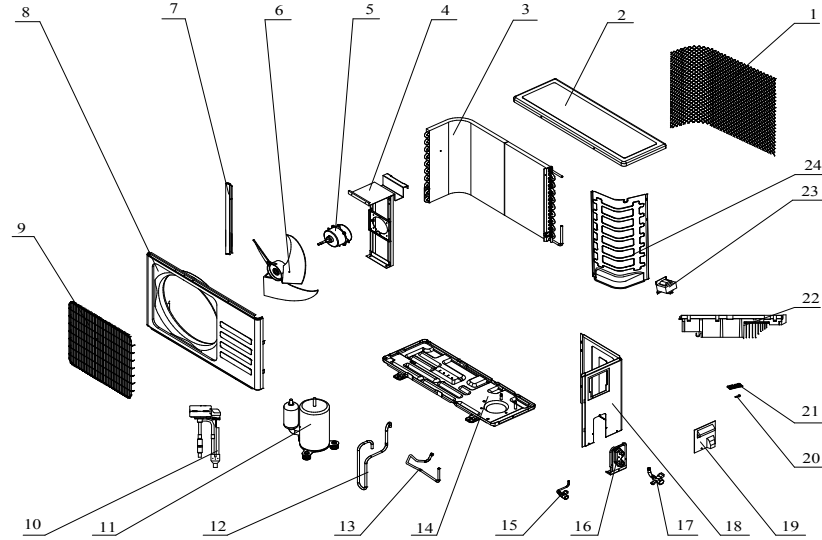
No.	Part No.	Part Name	Q'ty	Remark
1	42011-000184	Grille	1	
2	41207-000029	Top cover	1	
3	92011-000777	Condenser	1	
4	41203-000054	Outdoor motor supporter	1	
5	22001-000049_UM	Outdoor motor	1	
6	42004-000107	Propeller fan	1	
7	41205-000121	Left grille supporter	1	
8	41206-000052	Front plate	1	
9	42011-000038	Fan guard	1	
10	92014-000142	Compressor and accessories	1	
11	92008-000209	4-way valve	1	
12	92007-000969_UM	4-way valve assembly	1	
13	41202-000216	Base	1	
14	41205-000083	Right plate	1	
15	92007-001050	Two-way valve	1	
16	41204-000018	Valve supporter	1	
17	92007-001041_UM	Three-way valve	1	
18	41201-000044	Electronic Box Cover	1	
19	42001-000036	Cable clamp	1	
20	11304-100045_UM	Terminal	1	
21	31201-000988_UM	Outdoor PCB Assembly	1	
22	22011-000002_UM	Inductor	1	
23	92007-002083	Capillary assembly	1	
24	41208-000141	Partition plate	1	
25	10104-100047	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26	10104-100036	Discharge Temp. sensor	1	
27	41204-000051	Valve cover	1	
28	A2005-000530	Base carton	1	
29	A1201-007955	Cabinet carton	1	
30	A1202-000039	Base foaming	1	
31	41213-000030	Cover foaming	1	

Indoor Unit-MGTC1213E-20



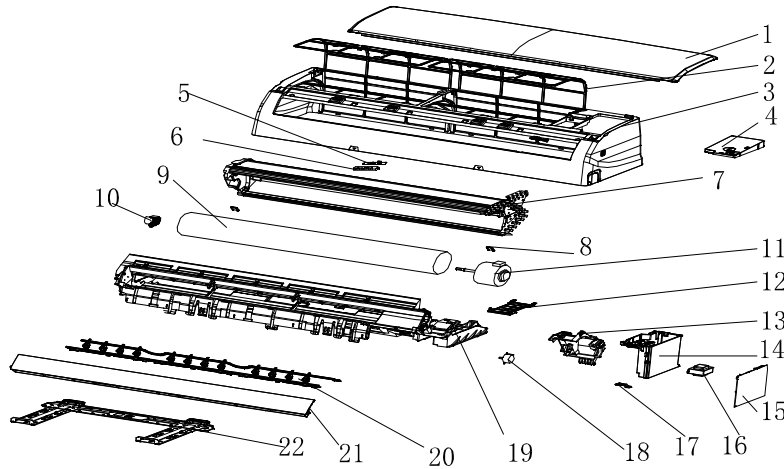
No.	Part No.	Part Name	Q'ty	Remark
1	41106-003591	Front Panel	1	
2	42008-000039	Air Filter	2	
3	41106-002375	Face Frame	1	
4	41211-000077	Electrical Box Cover	1	
5	31102-000084	Display PCB	1	
6	42003-000001	Display PCB Box	1	
7	92011-004784	Evaporator	1	
8	41108-000066	Screw Cover	2	
9	42004-000001	Cross Fan	1	
10	42007-000001	Bearing Mount	1	
11	22001-000307_UN	Indoor Motor	1	
12	41101-000242	In And Out Pipe Fixer	1	
13	42003-000051	Indoor Motor Cover	1	
14	41105-000134	Electrical Box	1	
15	31101-000272_UN	Main PCB	1	
16	NO	Transformer	1	
17	42001-000103	Cable Clamp	1	
18	22001-000313	Vane Motor 1	1	
19	41102-000091	Base	1	
20	41101-000082	Vertical Vane Assembly	2	
21	41103-000102	Vane	1	
22	41109-000041	Installation Plate	1	
23	10104-100014	Indoor Sensor Assembly	1	Not shown in Explosion view
24	22013-000026	Remote Controller	1	
25	32001-000057	wifi	1	
26	42009-000011	Drainage Hose	1	
27	A1101-007058	Indoor Carton	1	
28	41110-000217	Left Foaming	1	
29	41110-000262	Right Foaming	1	

Outdoor Unit- MGTC1213C-20



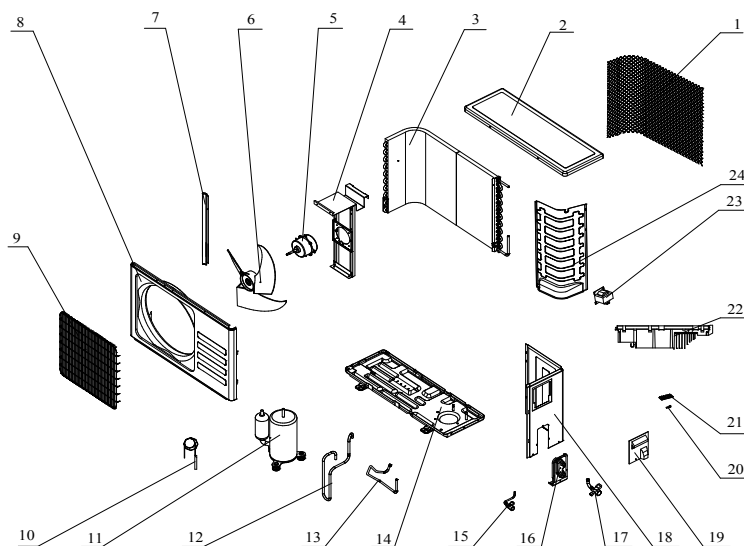
No.	Part No.	Part Name	Q'ty	Remark
1	42011-000186	Grille	1	
2	41207-000040	Top cover	1	
3	92011-004797	Condenser	1	
4	41203-000051	Outdoor motor supporter	1	
5	22001-000049_UN	Outdoor motor	1	
6	42004-000106	Propeller fan	1	
7	41205-000120	Left grille supporter	1	
8	41206-000059	Front plate	1	
9	42011-000041	Fan guard	1	
10	92007-002134	Capillary assembly	1	
11	92014-000142	Compressor	1	
12	92006-000507	Discharge Pipe	1	
13	92006-001664	Suction Pipe	1	
14	41202-000218	Base	1	
15	92007-001050	Two-way valve	1	
16	41204-000018	Valve supporter	1	
17	92007-001041	Three-way valve	1	
18	41205-000091	Right plate	1	
19	41201-000044	Electrical box cover	1	
20	42001-000036	Cable clamp	1	
21	11304-100046_UN	Terminal	1	
22	31201-001009_UN	Outdoor PCB assembly	1	
23	22011-000011_UN	Inductor	1	
24	41208-000147	Partition plate	1	
25	10104-100033	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26	10104-100036	Discharge Temp. sensor	1	
27	42001-000030	Capacitor Strip	1	
28	22003-000045_UN	Compressor Capacitor	1	
29	A1201-007900	Cabinet Carton	1	
30	A2005-000536	Base carton	1	
31	A1202-000041	Base foaming	1	
30	41213-000032	Cover foaming	1	

Indoor Unit- MGTC1223E-20



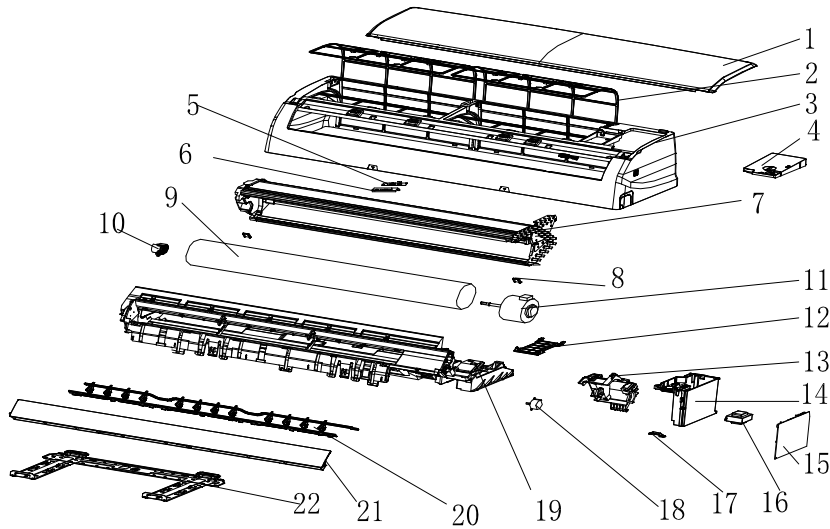
No.	Part No.	Part Name	Q'ty	Remark
1	41106-003591	Front Panel	1	
2	42008-000039	Air Filter	2	
3	41106-002375	Face Frame	1	
4	41211-000077	Electrical Box Cover	1	
5	31102-000084	Display PCB	1	
6	42003-000001	Display PCB Box	1	
7	92011-004784	Evaporator	1	
8	41108-000066	Screw Cover	2	
9	42004-000001	Cross Fan	1	
10	42007-000001	Bearing Mount	1	
11	22001-000267	Indoor Motor	1	
12	41101-000242	In And Out Pipe Fixer	1	
13	42003-000051	Indoor Motor Cover	1	
14	41105-000134	Electrical Box	1	
15	31101-000261	Main PCB	1	
16	NO	Transformer	1	
17	42001-000103	Cable Clamp	1	
18	22001-000313	Vane Motor 1	1	
19	41102-000091	Base	1	
20	41101-000082	Vertical Vane Assembly	2	
21	41103-000102	Vane	1	
22	41109-000041	Installation Plate	1	
23	10104-100014	Indoor Sensor Assembly	1	Not shown in Explosion view
24	22013-000946	Remote Controller	1	
25	32001-000047	WIFI	1	
26	42009-000011	Drainage Hose	1	
27	A1101-008531	Indoor Carton	1	
28	41110-000217	Left Foaming	1	
29	41110-000262	Right Foaming	1	

Outdoor Unit- MGTC1223C-20



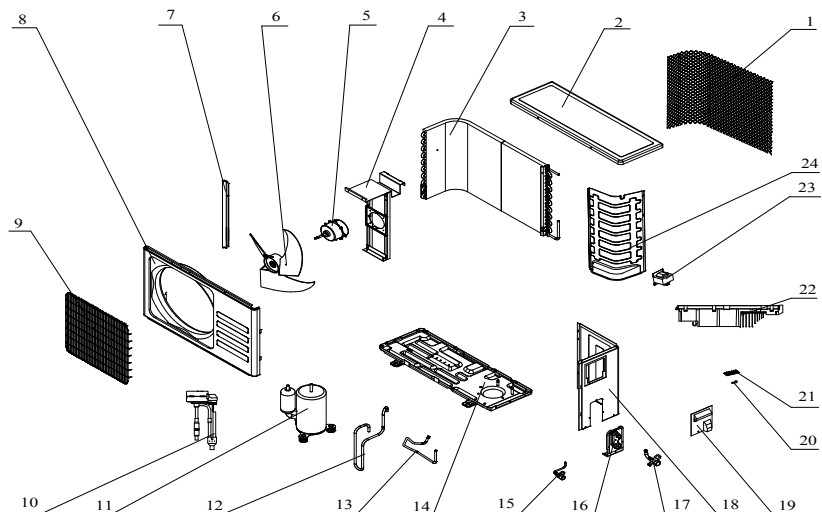
No.	Part No.	Part Name	Q'ty	Remark
1	42011-000183	Grille	1	Optional
2	41207-000029	Top Cover	1	
3	92011-004868	Condenser	1	
4	41203-000053	Outdoor Motor Supporter	1	
5	22001-000049	Outdoor Motor	1	
6	42004-000107	Propeller Fan	1	
7	41205-000121	Left Grille Supporter	1	
8	41206-000052	Front Plate	1	
9	42011-000038	Fan Guard	1	
10	92007-002134	Capillary Assembly	1	
11	92014-000142	Compressor	1	
12	92006-000478	Discharge pipe	1	
13	92006-005363	Suction pipe	1	
14	41202-000177	Base	1	
15	92007-001050	Two-way Valve	1	
16	41204-000018	Valve Supporter	1	
17	92007-001041	Three-way Valve	1	
18	41205-000083	Right Plate	1	
19	41201-000044	Electrical Box Cover	1	
20	42001-000036	Cable Clamp	1	
21	11304-100045	terminal	1	
22	31201-001063	Outdoor PCB assembly	1	
23	22011-000002	Inductor	1	
24	41208-000141	Partition plate	1	
25	10104-100047	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26	10104-100036	Discharge Temp. sensor	1	
27	A1201-009174	Cabinet Carton	1	
28	A2005-000530	Base carton	1	
29	A1202-000039	Base foaming	1	
30	41213-000030	Cover foaming	1	

Indoor Unit- MGTC1823E-20



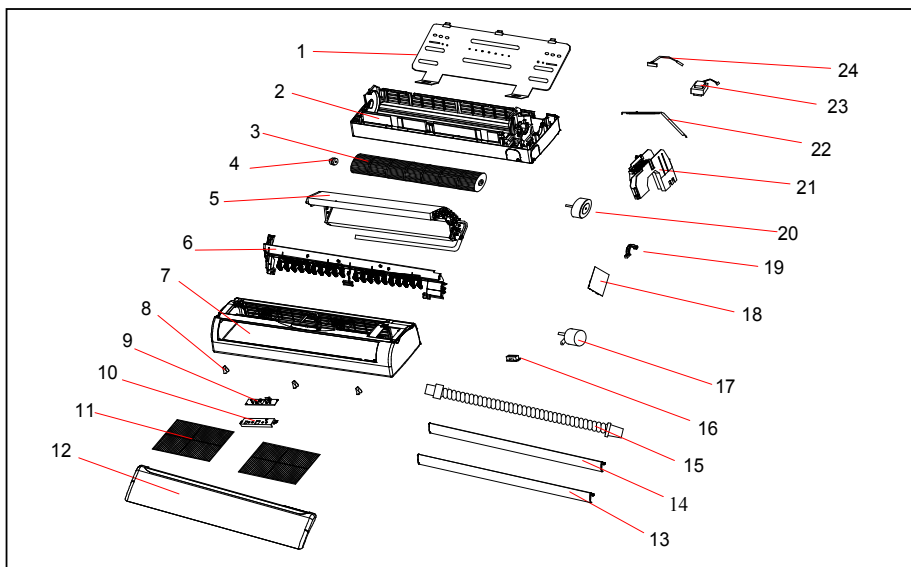
No.	Part No.	Part Name	Q'ty	Remark
1	41106-002202	Front Panel	1	
2	42008-000001	Air Filter	2	
3	41106-002351	Face Frame	1	
4	41211-000079	Electrical Box Cover	1	
5	31102-000084	Display PCB	1	
6	42003-000001	Display PCB Box	1	
7	92011-004728	Evaporator	1	
8	41108-000066	Screw Cover	2	
9	42004-000002	Cross Fan	1	
10	42007-000001	Bearing Mount	1	
11	22001-000257	Indoor Motor	1	
12	41101-000224	In And Out Pipe Fixer	1	
13	42003-000045	Indoor Motor Cover	1	
14	41105-000134	Electrical Box	1	
15	31101-000273	Main PCB	1	
16	NO	Transformer	1	
17	42001-000103	Cable Clamp	1	
18	22001-000321	Vane Motor 1	1	
19	41102-000076	Base	1	
20	41101-000042	Vertical Vane Assembly	3	
21	41103-000142	Vane	1	
22	41109-000019	Installation Plate	1	
23	10104-100014	Indoor Sensor Assembly	1	Not shown in Explosion view
24	22013-000946	Remote Controller	1	
25	42009-000011	Drainage Hose	1	
26	32001-000033	WIFI	1	
27	81102-002993	Indoor Carton	1	
28	A2006-000070	Left Foaming	1	
29	A2006-000068	Right Foaming	1	
30	42005-000069	Middle support	1	

Outdoor Unit-MGTC1823C-20



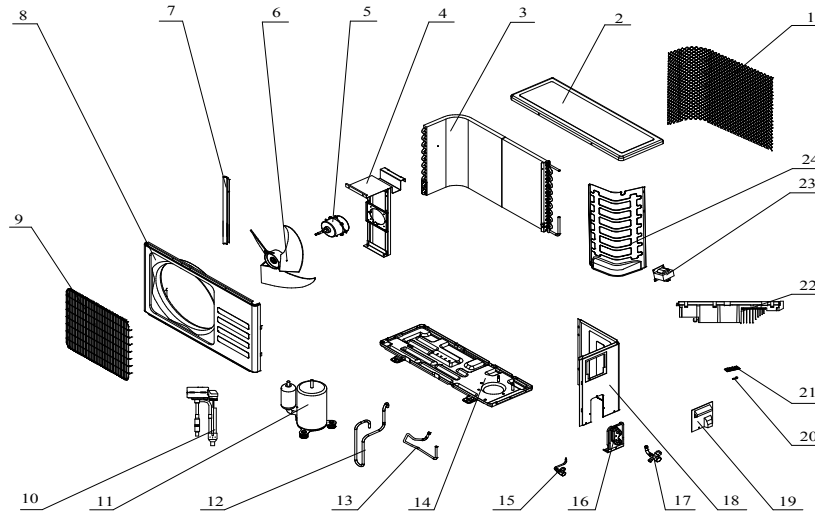
No.	Part No.	Part Name	Q'ty	Remark
1	42011-000169	Grille	1	
2	41207-000033	Top cover	1	
3	92011-000616	Condenser	1	
4	41203-000052	Outdoor motor supporter	1	
5	22001-000096	Outdoor motor	1	
6	42004-000104	Propeller fan	1	
7	41205-000119	Left grille supporter	1	
8	41206-000057	Front plate	1	
9	42011-000040	Fan guard	1	
10	92007-002228	Capillary assembly	1	
11	92014-000318	Compressor	1	
12	92006-000489	Discharge Pipe	1	
13	92006-001654	Suction Pipe	1	
14	41202-000156	Base	1	
15	92007-001051	Two-way valve	1	
16	41204-000018	Valve supporter	1	
17	92007-001041	Three-way valve	1	
18	41205-000090	Right plate	1	
19	41201-000044	Electrical box cover	1	
20	42001-000036	Cable clamp	1	
21	11304-100045	Terminal	1	
22	31201-001025	Outdoor PCB assembly	1	
23	22011-000014	Inductor	1	
24	41208-000154	Partition plate	1	
25	10104-100038	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26	10104-100036	Discharge Temp. sensor	1	
27	A1201-007956	Cabinet Carton	1	
28	A2005-000532	Base carton	1	
29	A1202-000037	Base foaming	1	
30	41213-000031	Cover foaming	1	

Indoor Unit- MGTC2423E-20



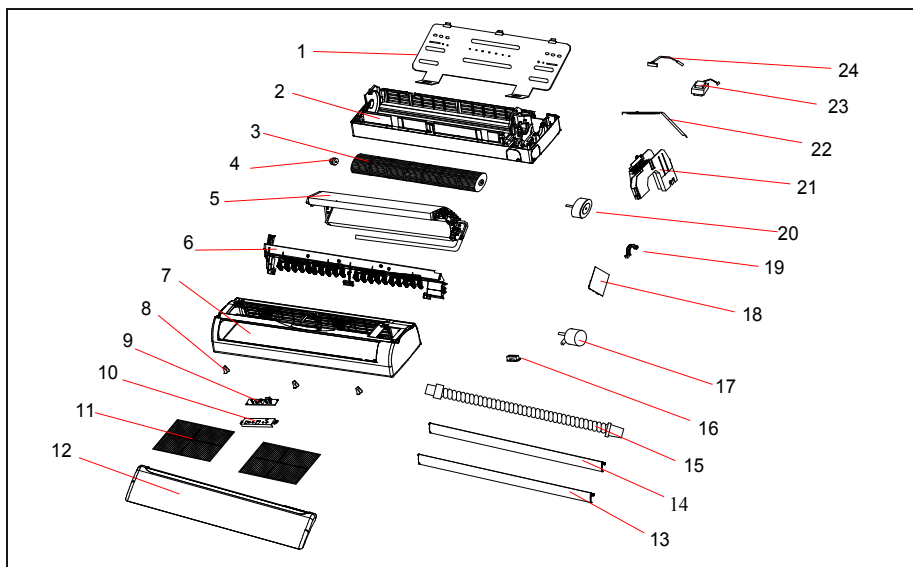
No.	Part No.	Part Name	Q'ty	Remark
1	41109-000028	Installation Plate	1	
2	41199-002746	Base	1	
3	42004-000042	Cross Fan	1	
4	42007-000001	Bearing Mount	1	
5	92011-004785	Evaporator	1	
6	41103-000064	Water Drainage Assembly	1	
7	41106-002275	Face Frame	1	
8	41108-000066	Screw Cover	3	
9	31102-000085	Display PCB	1	
10	42003-000001	Display PCB Box	1	
11	42008-000041	Air Filter	2	
12	41106-002203	Front Panel	1	
13	41103-000317	Up Vane	1	
14	41103-000312	Down Vane	1	
15	42009-000004	Drainage Hose	1	
16	42001-000103	Cable Clamp	1	
17	22001-000357	Vane Motor	2	
18	31101-000256	Main PCB	1	
19	41108-000006	Indoor Motor Cover	1	
20	22001-000240	Indoor Motor	1	
21	41105-000139	Electrical Box	1	
22	41211-000080	Electrical Box Cover	1	
23	NO	Transformer	1	
24	10104-100022	Indoor Sensor Assembly	1	
25	22013-000946	Remote Controller	1	Not shown in Explosion view
26	32001-000033	WIFI	1	
27	A1101-007184	Indoor Carton	1	
28	A2004-000284	Left Foaming	1	
29	A2004-000332	Right Foaming	1	
30	A1101-005787	Middle Foaming Supporter	1	

Outdoor Unit: MGTC2423C-20



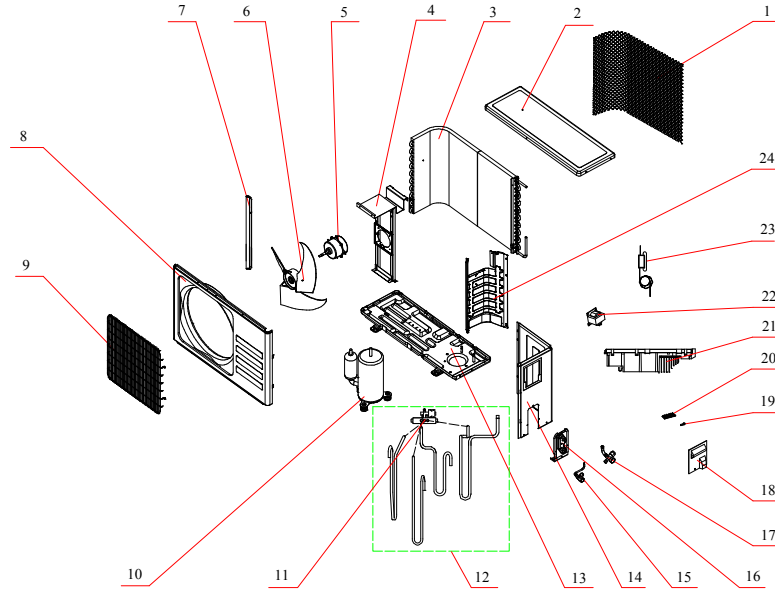
No.	Part No.	Part Name	Q'ty	Remark
1	42011-000188	Grille	1	
2	41207-000028	Top cover	1	
3	92011-004798	Condenser	1	
4	41203-000049	Outdoor motor supporter	1	
5	22001-000110	Outdoor motor	1	
6	42004-000105	Propeller fan	1	
7	41205-000117	Left grille supporter	1	
8	41206-000058	Front plate	1	
9	42011-000039	Fan guard	1	
10	92007-002133	Capillary assembly	1	
11	92014-000318	Compressor	1	
12	92006-000381	Discharge Pipe	1	
13	92007-005313	Suction Pipe	1	
14	41202-000186	Base	1	
15	92007-001052	Two-way valve	1	
16	41204-000018	Valve supporter	1	
17	92007-001042	Three-way valve	1	
18	41205-000088	Right plate	1	
19	41201-000044	Electrical box cover	1	
20	42001-000036	Cable clamp	1	
21	11304-100045	Terminal	1	
22	31201-001013	Outdoor PCB assembly	1	
23	22011-000004	Inductor	1	
24	41208-000155	Partition plate	1	
25	10104-100030	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in explosion view
26	10104-100037	Discharge Temp. sensor	1	
27	A1201-007784	Cabinet Carton	1	
28	A2005-000521	Base carton	1	
29	A1202-000030	Base foaming	1	
30	41110-000204	Cover foaming	1	

Indoor Unit- MGTC3623E-17



No.	Part No.	Part Name	Q'ty	Remark
1	41109-000028	Installation Plate	1	
2	41199-002746	Base	1	
3	42004-000042	Cross Fan	1	
4	42007-000001	Bearing Mount	1	
5	92011-004396	Evaporator	1	
6	41103-000064	Water Drainage Assembly	1	
7	41106-002275	Face Frame	1	
8	41108-000066	Screw Cover	3	
9	31102-000085	Display PCB	1	
10	42003-000001	Display PCB Box	1	
11	42008-000041	Air Filter	2	
12	41106-002203	Front Panel	1	
13	41103-000317	Up Vane	1	
14	41103-000312	Down Vane	1	
15	42009-000004	Drainage Hose	1	
16	42001-000103	Cable Clamp	1	
17	22001-000357	Vane Motor	2	
18	31101-000267	Main PCB	1	
19	41108-000006	Indoor Motor Cover	1	
20	22001-000240	Indoor Motor	1	
21	41105-000139	Electrical Box	1	
22	41211-000080	Electrical Box Cover	1	
23	NO	Transformer	1	
24	10104-100022	Indoor Sensor Assembly	1	
25	22013-000946	Remote Controller	1	Not shown in Explosion view
26	32001-000057	WIFI	1	
27	A1101-007286	Indoor Carton	1	
28	A2004-000284	Left Foaming	1	
29	A2004-000332	Right Foaming	1	
30	A1103-000012	Middle Foaming Supporter	1	

Outdoor Unit- MGTC3623C-17



No.	Part No.	Part Name	Q'ty	Remark
1	42011-000166	Back Grille	1	Optional
2	41207-000016	Top cover	1	
3	92011-004586	Condenser	1	
4	41203-000030	Outdoor motor supporter	1	
5	22001-000120	Outdoor motor	1	
6	42004-000091	Propeller fan	1	
7	41205-000125	Left grille supporter	1	
8	41206-000046	Front plate	1	
9	42011-000031	Fan guard	1	
10	92014-000300	Compressor	1	
11	92008-000206	4-way Valve	1	
12	92007-000984	4-way Valve Assembly	1	
13	41202-000229	base	1	
14	41205-000064	Right plate	1	
15	92008-000047	Two-way valve	1	
16	41204-000011	Valve supporter	1	
17	92008-000119	Three-way valve	1	
18	41201-000043	Electrical box cover	1	
19	42001-000036	Cable clamp	1	
	42001-000038	Cable clamp	1	
20	11304-100045	Terminal	1	
21	31201-000981	Outdoor PCB assembly	1	
22	22011-000004	Inductor	1	
23	92007-002096	Capillary Assembly	1	
24	41208-000176	Partition plate	1	
25	10104-100039	Pipe Temp. sensor and outdoor Temp. sensor	1	Not shown in Explosion view
26	10104-100037	Discharge Temp. sensor	1	
27	41204-000047	Valve cover	1	
28	A2005-000509	Base carton	1	
29	A1201-007893	Cabinet carton	1	
30	A1202-000022	Base foaming	1	
31	41213-000017	Cover foaming	1	

7. Precaution

7.1 SAFETY RULES AND RECOMMENDATIONS FOR THE INSTALLATION

- Read this guide before installing and using the appliance.
- During the installation of the indoor and outdoor units the access to the working area should be forbidden to children.
Unforeseeable accidents could happen.
- Make sure that the base of the outdoor unit is firmly fixed.
- Check that air cannot enter the refrigerant system and check for refrigerant leaks when moving the air conditioner.
- Check that air cannot enter the refrigerant system and check for refrigerant leaks when moving the air conditioner.
- The ratings of the fuse installed in the built in-control unit are T 5A / 250V.
- The user must protect the indoor unit with a fuse of suitable capacity for the maximum input current or with another overload protection device.
- Ensure that the mains voltage corresponds to that stamped on the rating plate. Keep the switch or power plug clean. Insert the power plug correctly and firmly into the socket, thereby avoiding the risk of electric shock or fire due to insufficient contact.
- Check that the socket is suitable for the plug, otherwise have the socket changed.
- The appliance must be fitted with means for disconnection from the supply mains having a contact separation in all poles that provide full disconnection under over voltage category III conditions, and these means must be incorporated in the fixed wiring in accordance with the wiring rules.
- The air conditioner must be installed by professional or qualified persons. Do not install the appliance at a distance of less than 50 cm from inflammable substances (alcohol, etc.) Or from pressurized containers (e.g. spray cans).
- If the appliance is used in areas without the possibility of ventilation, precautions must be taken to prevent any leaks of refrigerant gas from remaining in the environment and creating a danger of fire.
- The packaging materials are recyclable and should be disposed of in the separate waste bins. Take the air conditioner at the end of its useful life to a special waste collection center for disposal.
- Only use the air conditioner as instructed in this booklet. These instructions are not intended to cover every possible condition and situation. As with any electrical household appliance, common sense and caution are therefore always recommended for installation, operation and maintenance.
- The appliance must be installed in accordance with applicable national regulations.
- Before accessing the terminals, all the power circuits must be disconnected from the power supply.
- The appliance shall be installed in accordance with national wiring regulations.
- This appliance can be used by children aged from 8 years and above and persons with reduced physical, sensory or mental capabilities or lack of experience and knowledge if they have been given supervision or

instruction concerning use of the appliance in a safe way and understand the hazards involved. Children shall not play with the appliance. Cleaning and user maintenance shall not be made by children without supervision.

7.2 SAFETY RULES AND RECOMMENDATIONS FOR THE USER

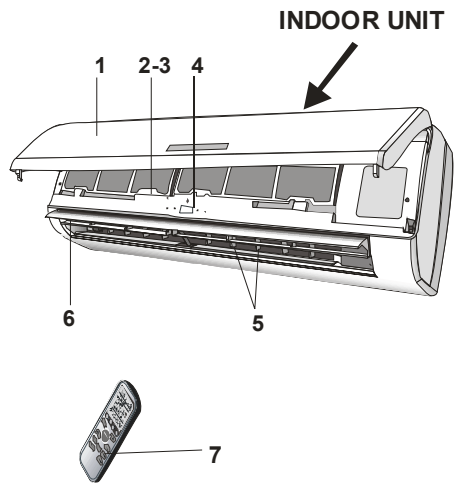
- Do not try to install the conditioner alone; always contact specialized technical personnel. Cleaning and maintenance must be carried out by specialized technical personnel. In any case disconnect the appliance from the mains electricity supply before carrying out any cleaning or maintenance.
- Ensure that the mains voltage corresponds to that stamped on the rating plate. Keep the switch or power plug clean. Insert the power plug correctly and firmly into the socket, thereby avoiding the risk of electric shock or fire due to insufficient contact.
- Do not pull out the plug to switch off the appliance when it is in operation, since this could create a spark and cause a fire, etc.
- This appliance has been made for air conditioning domestic environments and must not be used for any other purpose, such as for drying clothes, cooling food, etc.
- The packaging materials are recyclable and should be disposed of in the separate waste bins. Take the air conditioner at the end of its useful life to a special waste collection center for disposal.
- Always use the appliance with the air filter mounted. The use of the conditioner without air filter could cause an excessive accumulation of dust or waste on the inner parts of the device with possible subsequent failures.
- The user is responsible for having the appliance installed by a qualified technician, who must check that it is earthed in accordance with current legislation and insert a thermomagnetic circuit breaker.
- The batteries in remote controller must be recycled or disposed of properly. Disposal of Scrap Batteries
--- Please discard the batteries as sorted municipal waste at the accessible collection point.
- Never remain directly exposed to the flow of cold air for a long time. The direct and prolonged exposition to cold air could be dangerous for your health. Particular care should be taken in the rooms where there are children, old or sick people.
- If the appliance gives off smoke or there is a smell of burning, immediately cut off the power supply and contact the Service Centre.
- The prolonged use of the device in such conditions could cause fire or electrocution.
- Have repairs carried out only by an authoritative Service Centre of the manufacturer. Incorrect repair could expose the user to the risk of electric shock, etc.
- Unhook the automatic switch if you foresee not to use the device for a long time.
The airflow direction must be properly adjusted.
- The flaps must be directed downwards in the heating mode and upwards in the cooling mode.
- Only use the air conditioner as instructed in this booklet. These instructions are not intended to cover every possible condition and situation. As with any electrical household appliance, common sense and caution are therefore always recommended for installation, operation and maintenance.

- Ensure that the appliance is disconnected from the power supply when it will remain inoperative for a long period and before carrying out any cleaning or maintenance.
- Selecting the most suitable temperature can prevent damage to the appliance.

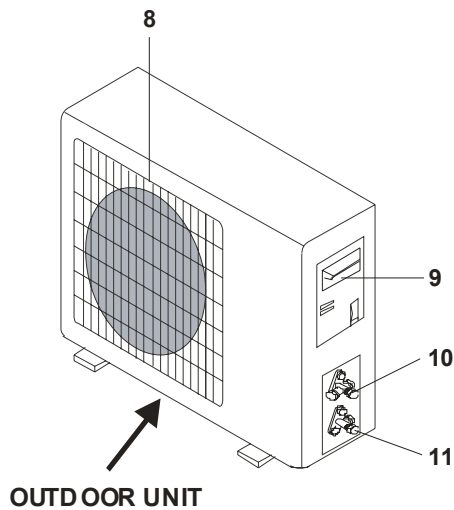
7.3 SAFETY RULES AND PROHIBITIONS

- Do not bend, tug or compress the power cord since this could damage it. Electrical shocks or fire are probably due to a damaged power cord. Specialized technical personnel only must replace a damaged power cord.
- Do not use extensions or gang modules.
- Do not touch the appliance when barefoot or parts of the body are wet or damp.
- Do not obstruct the air inlet or outlet of the indoor or the outdoor unit. The obstruction of these openings causes a reduction in the operative efficiency of the conditioner with possible consequent failures or damages.
- In no way alter the characteristics of the appliance.
- Do not install the appliance in environments where the air could contain gas, oil or sulphur or near sources of heat.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- Do not climb onto or place any heavy or hot objects on top of the appliance.
- Do not leave windows or doors open for long when the air conditioner is operating.
- Do not direct the airflow onto plants or animals.
- A long direct exposition to the flow of cold air of the conditioner could have negative effects on plants and animals.
- Do not put the conditioner in contact with water. The electrical insulation could be damaged and thus causing electrocution.
- Do not climb onto or place any objects on the outdoor unit
- Never insert a stick or similar object into the appliance. It could cause injury.
- Children should be supervised to ensure that they do not play with the appliance. If the supply cord is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard.

8.Names of parts



Indoor unit	
No.	Name
1	Front panel
2	Air filter
3	Special filter(option)
4	Display PCB
5	Vertical vane
6	Horizontal vane
7	Remote controller



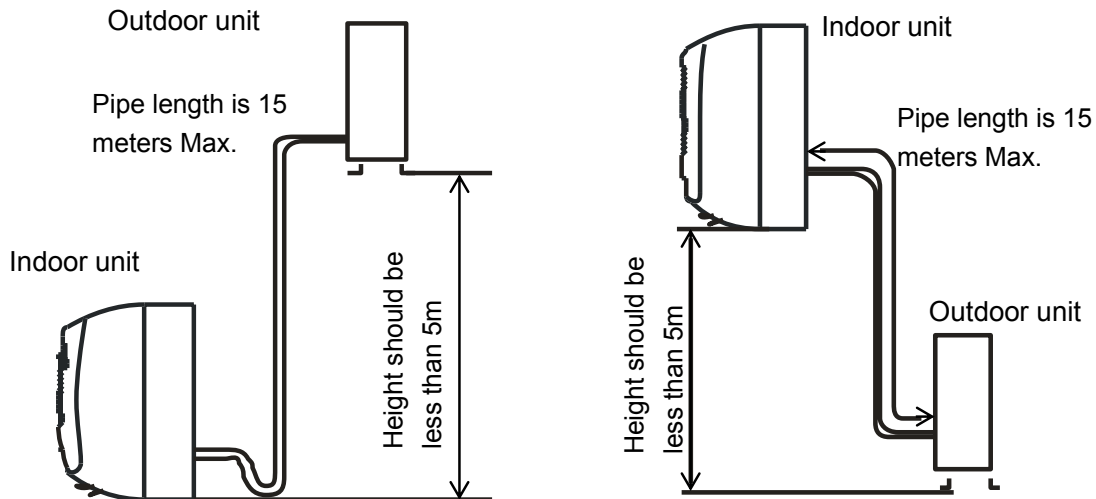
Outdoor unit	
No.	Name
8	Air outlet grille
9	Electronic box cover
10	2-way valve
11	3-way valve

9. Installation manual

9.1 Installation Details

Connecting pipe length

Model	Pipe size(Inch)	
	Liquid	Gas
MGTC0913E-20 /MGTC0913C-20	1/4	3/8
MGTC0923E-20 /MGTC0923C-20	1/4	3/8
MGTC1213E-20 /MGTC1213C-20	1/4	3/8
MGTC1223E-20 /MGTC1223C-20	1/4	3/8
MGTC1823E-20 /MGTC1823C-20	1/4	3/8
MGTC2423E-20 /MGTC2423C-20	1/4	1/2
MGTC3623E-17 /MGTC3623C-17	1/4	5/8



Mode	Standard length: m	Refrigerant piping Max. length: m A	Additional refrigerant B Calculation: $\times g = Bg/m(A-5m)$
MGTC0913E-20 /MGTC0913C-20	3	15	20g/m
MGTC0923E-20 /MGTC0923C-20	3	15	20g/m
MGTC1213E-20 /MGTC1213C-20	3	15	20g/m
MGTC1223E-20 /MGTC1223C-20	3	15	20g/m
MGTC1823E-20 /MGTC1823C-20	4	15	30g/m
MGTC2423E-20 /MGTC2423C-20	4	15	30g/m
MGTC3623E-17 /MGTC3623C-17	4	15	30g/m

Connecting cables

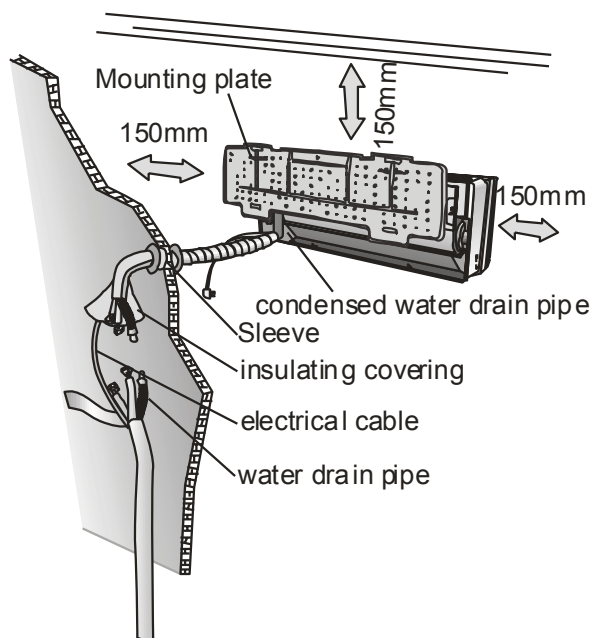
The power cord should be selected according to the following specifications sheet.

Appliance Amps	Wire Size
5	AWG21/0.75 mm ²
10	AWG18/1.0 mm ²
13	AWG15 /1. 5 mm ²
18	AWG14/1.6 mm ²
25	AWG12/2.0 mm ²
30	AWG10/2.5 mm ²

9.2 Installation for the first time

Indoor unit

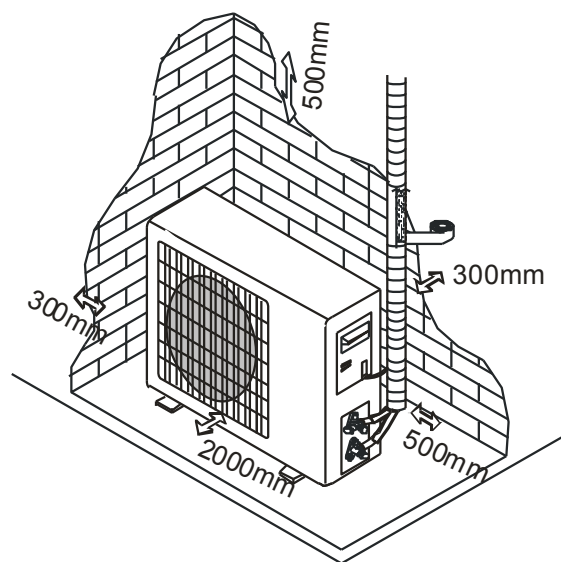
- Install the indoor unit level on a strong wall that is not subject to vibrations
- The inlet and outlet ports should not be obstructed: the air should be able to blow all over the room.
- Do not install the unit near a source of heat, steam, or flammable gas.
- Install the unit near an electric socket or private circuit.
- Do not install the unit where it will be exposed to direct sunlight.
- Install the unit where connection between indoor and outdoor unit is as easy as possible.
- Install the unit where it is easy to drain the condensed water.
- Check the machine operation regularly and leave the necessary spaces as shown in the picture.
- Install the indoor unit where the filter can be easily accessible.



OUTDOOR UNIT

- Do not install the outdoor unit near sources of heat, steam or flammable gas.
- Do not install the unit in too windy or dusty places.
- Do not install the unit where people often pass. Select a place where the air discharge and operating sound level will not disturb the neighbours.
- Avoid installing the unit where it will be exposed to direct sunlight (other wise use a protection, if necessary, that should not interfere with the air flow).
- Leave the spaces as shown in the picture for
- The air to circulate freely.
- Install the outdoor unit in a safe and solid place.
- If the outdoor unit is subject to vibration, place rubber gaskets onto the feet of the unit.
- Install the indoor unit in the room to be air conditioning, avoiding to installation in corridors or communal areas.

Minimum space to be left (mm) showing in the picture.

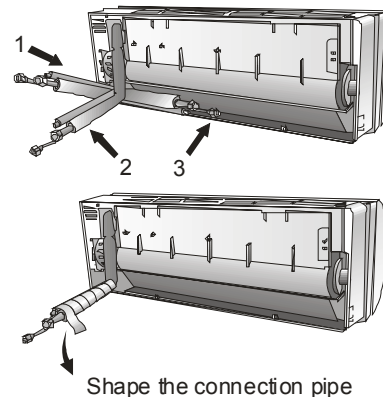


- 7).An efficient earth connection must be ensured.
- 8).If the power cable is damaged, it must be replaced by an authorized Service Centre.

9.2.4 Refrigerant piping connection

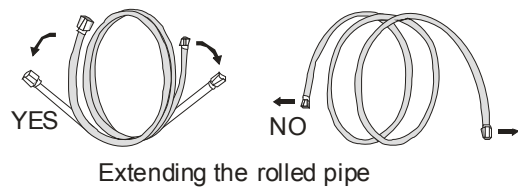
The piping can be run in the 3 directions indicated by numbers in the picture. When the piping is run in direction 1 or 3, cut a notch along the groove on the side of the indoor unit with a cutter.

Run the piping in the direction of the wall hole and bind the copper pipes, the drain pipe and the power cables together with the tape with the drain pipe at the bottom, so that water can flow freely.



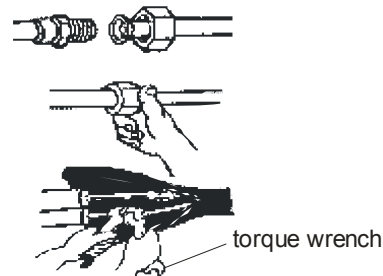
9.2.5 Connecting the pipes.

- Do not remove the cap from the pipe until connecting it, to avoid dampness or dirt from entering.
- If the pipe is bent or pulled too often, it will become stiff. Do not bend the pipe more than three times at one point.
- When extending the rolled pipe, straighten the pipe by unwinding it gently as shown in the picture.



9.2.6 Connections to the indoor unit

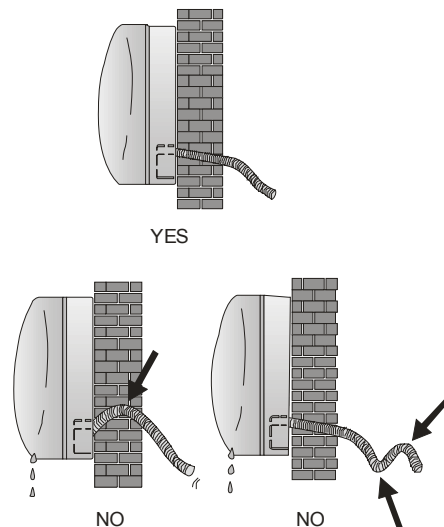
- 1).Remove the indoor unit pipe cap (check that there is no debris inside).
- 2).Insert the flare nut and create a flange at the extreme end of the connection pipe.
- 3).Tighten the connections by using two wrenches working in opposite directions.



9.2.7 Indoor unit condensed water drainage

The indoor unit condensed water drainage is fundamental for the success of the installation.

- 1).Place the drain hose below the piping, taking care not to create siphons.
- 2).The drain hose must slant downwards to aid drainage.
- 3).Do not bend the drain hose or leave it protruding or twisted and do not put the end of it in water. If an extension is connected to the drain hose, ensure that it is lagged when it passes into the indoor unit.

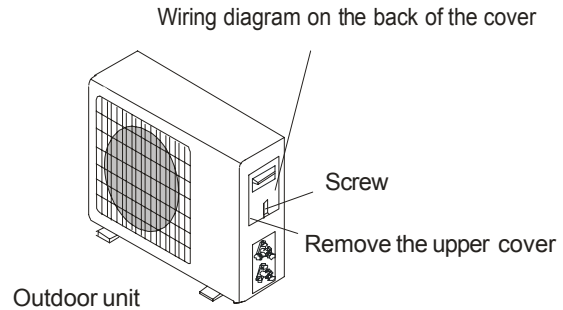


- 4). If the piping is installed to the right, the pipes, power cable and drain hose must be lagged and secured onto the rear of the unit with a pipe connection.

- Insert the pipe connection into the relative slot.
- Press to join the pipe connection to the base.

9.2.8 Electronic connections

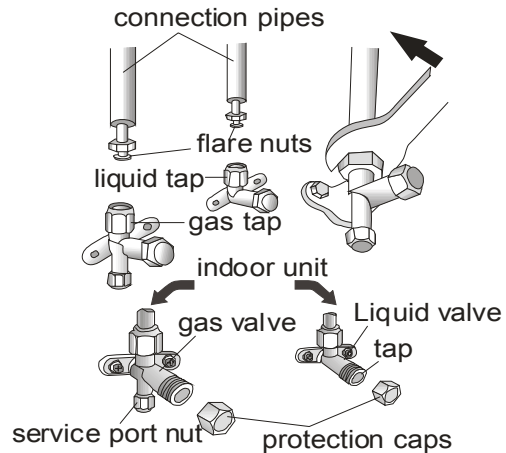
1. Take the cover away.
2. Connect the cable wires to the terminal board using the same numbering as in the indoor unit.
3. For the electrical connections, see the wiring diagram on the back of the cover
4. Fasten the cables with a cable-clamp.
5. An efficient earth connection must be ensured.
6. Replace the covers.



9.2.9 Connecting the pipe

Screw the flare nuts to the outdoor unit coupling with the same tightening procedures described for the indoor unit.

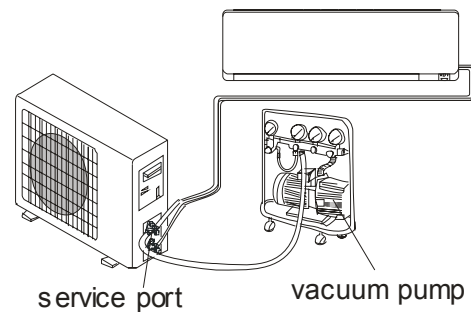
Note: If the tightening torque is not sufficient, there will probably be some leakage. With excessive tightening torque there will also be some leakage, as the flange could be damaged.



9.2.10 Bleeding

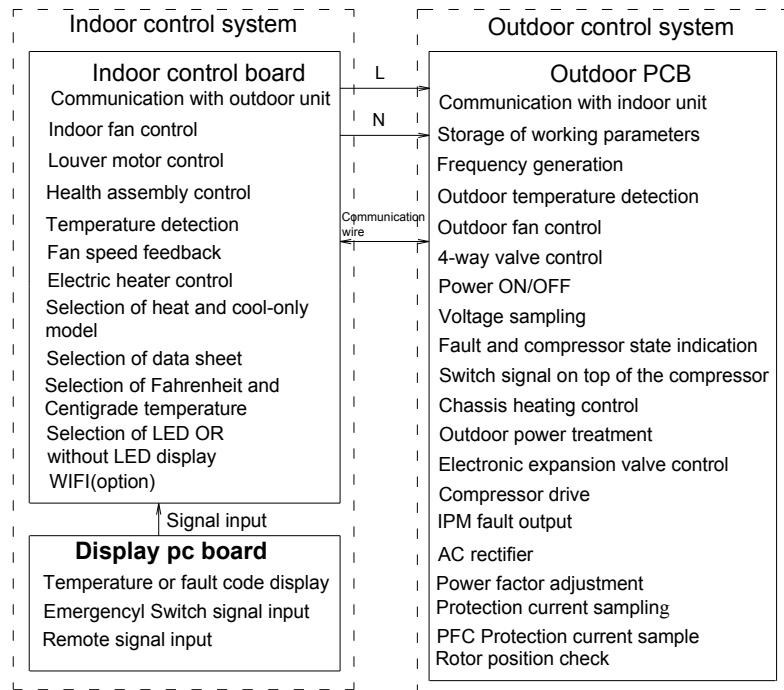
Air and humidity left inside the refrigerant circuit can cause compressor malfunction. After having connected the indoor and outdoor units, bleed the air and humidity from the refrigerant circuit by using a vacuum pump.

The air and humidity left inside the refrigerant circulation can cause compressor malfunction. After having connected the indoor and outdoor units, bleed the air and humidity from the refrigerant circulation using a vacuum pump.



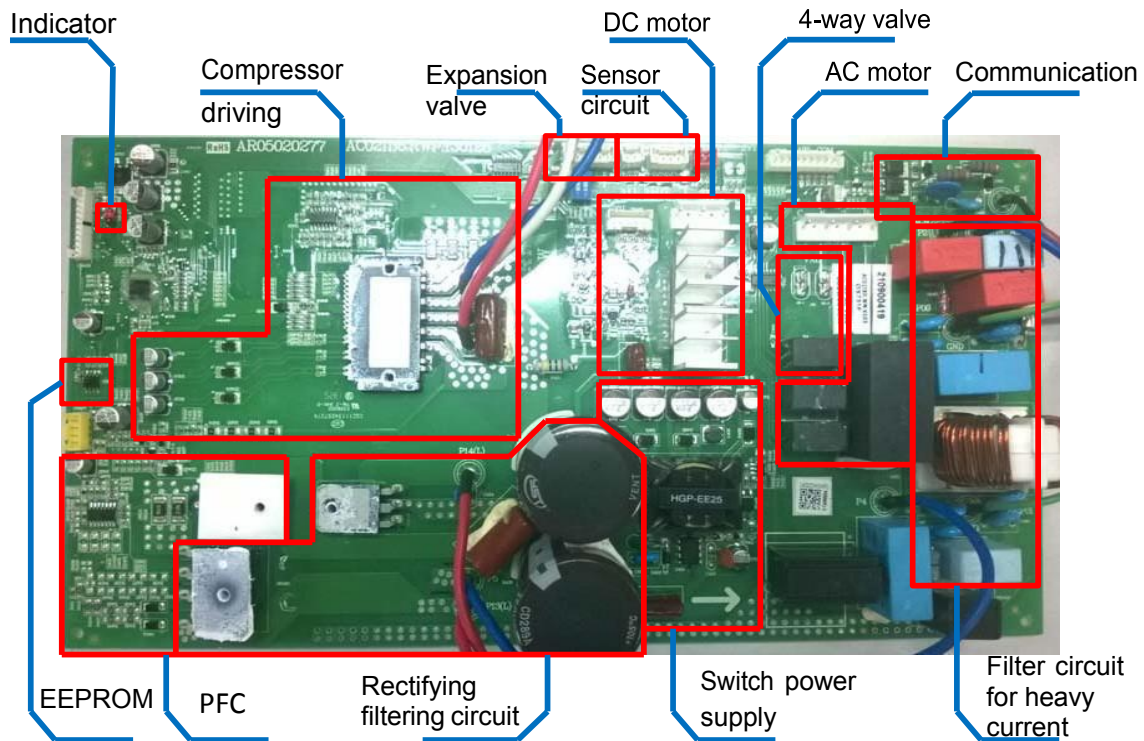
10. Trouble shooting

10.1 Outdoor control diagram

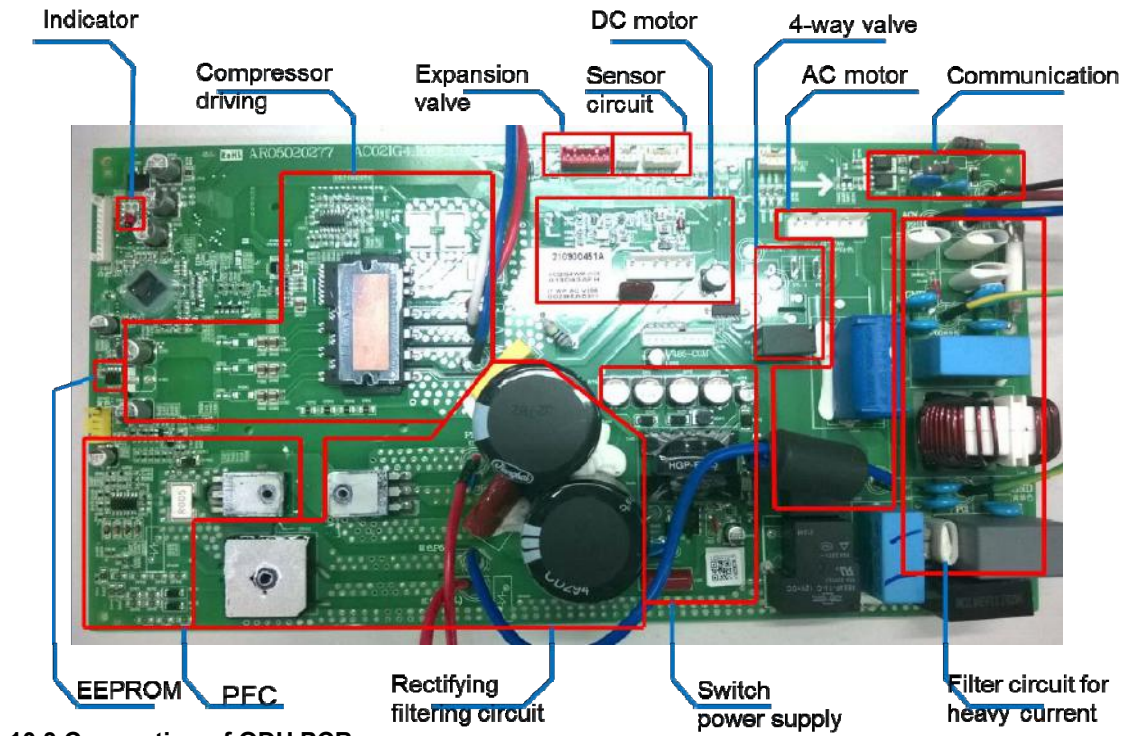


10.2 The structure of ODU PCB

09K,12K

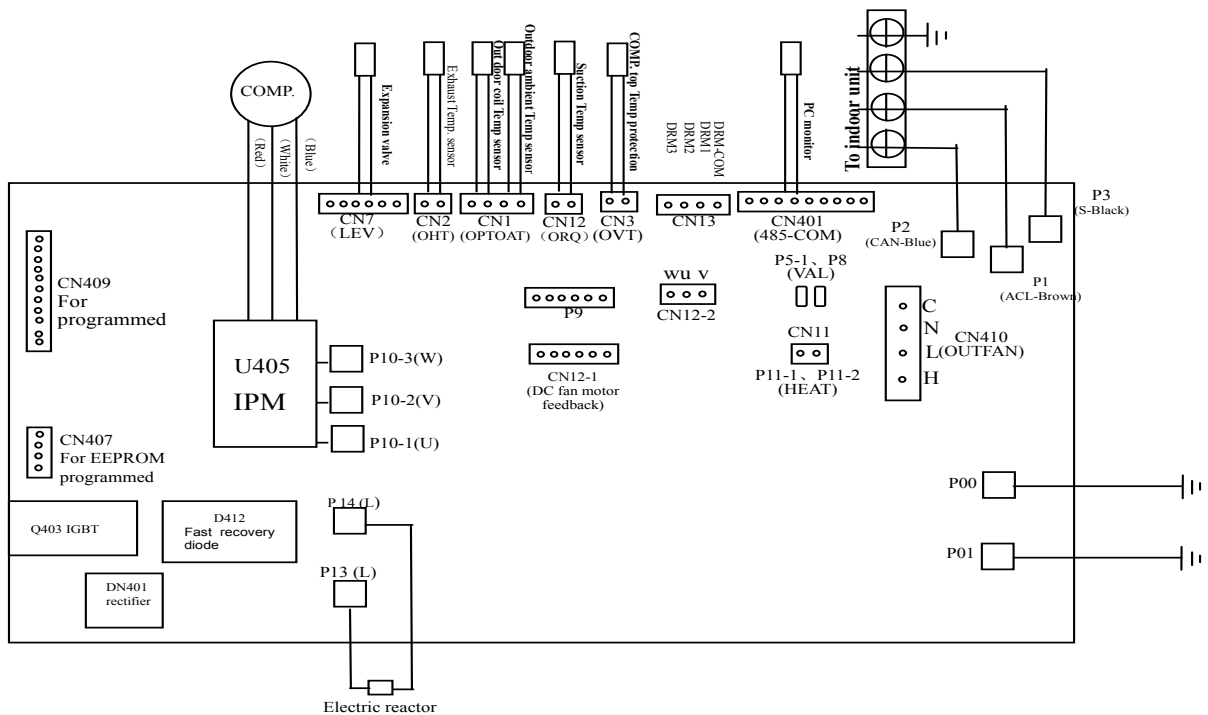


18K,24K

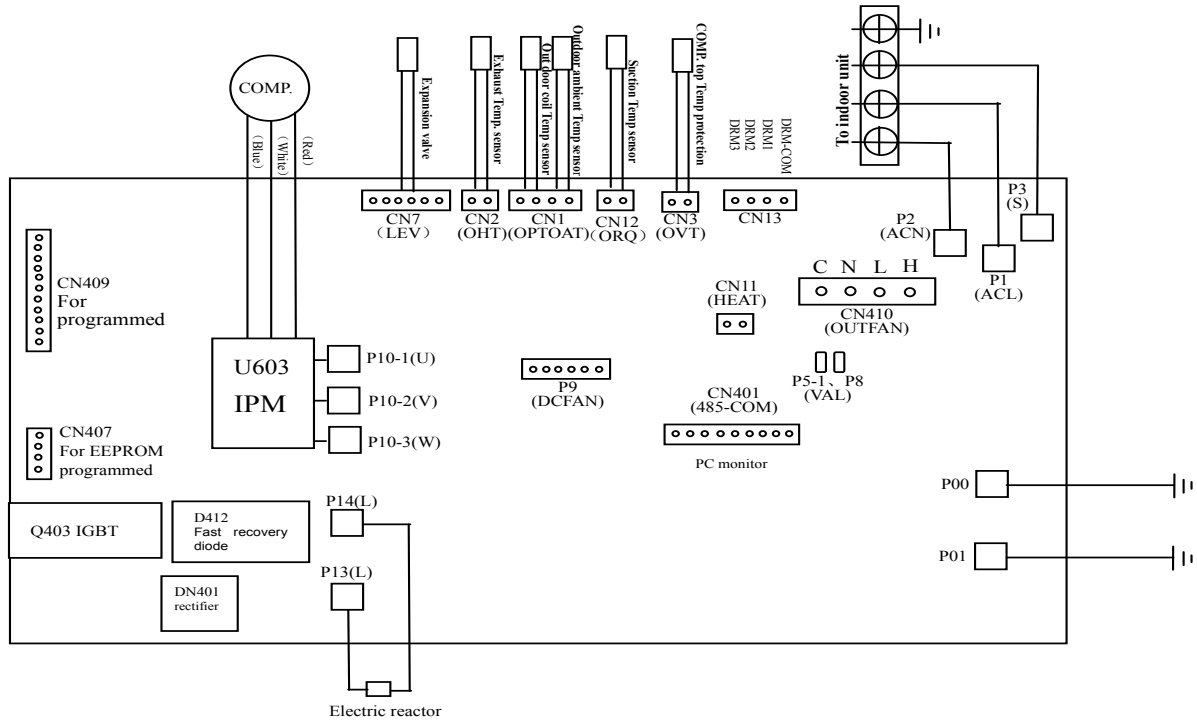


10.3 Connection of ODU PCB

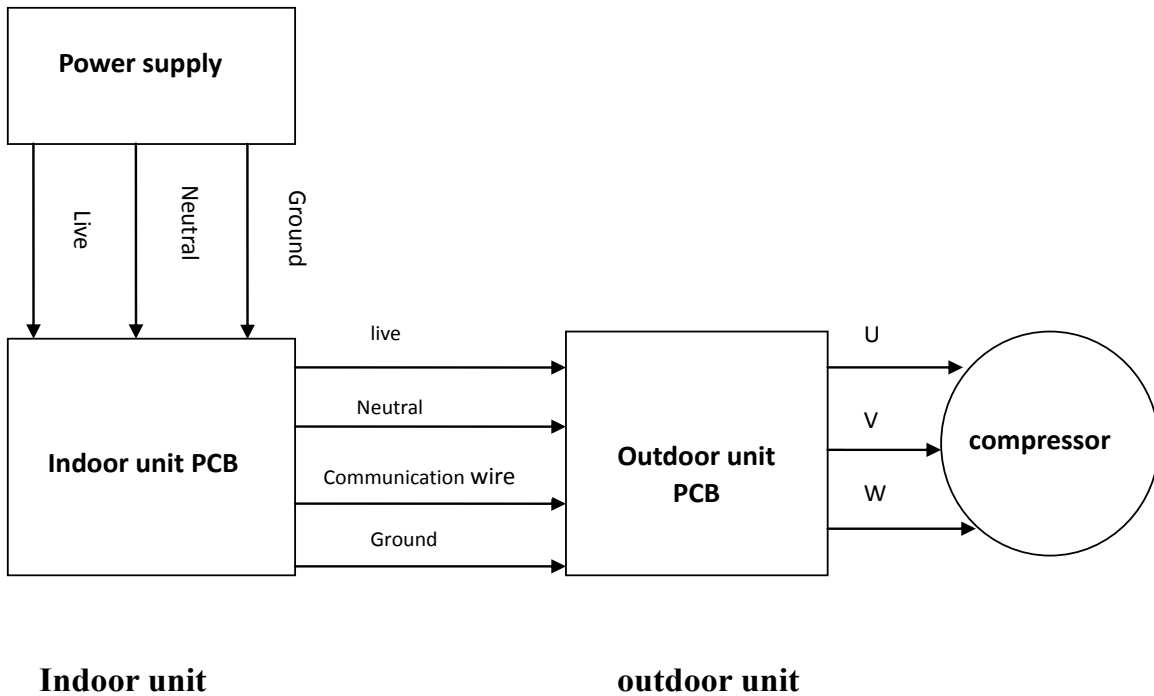
09K,12K



18K,24K



10.4 Current flow

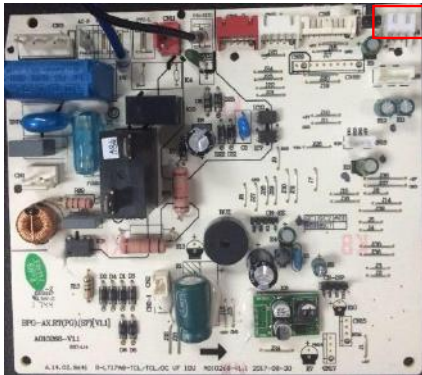


10.5 Examples of repairing

10.5.1 Display E1 or E2

Reasons:

- 1) The sensor connection terminal loose or not plugged in.



Solution: Check the connector of sensor and slot (CN6), if loose or not plugged in, please connect again.

- 2) Room temperature sensor (IRT) and Indoor pipe (coil) temperature sensor (IPT) damage (short or broken).

Solution: check the resistance of the sensor $R(25^{\circ}\text{C}) = 5\text{ k}\Omega$, if short or broken please replace it.



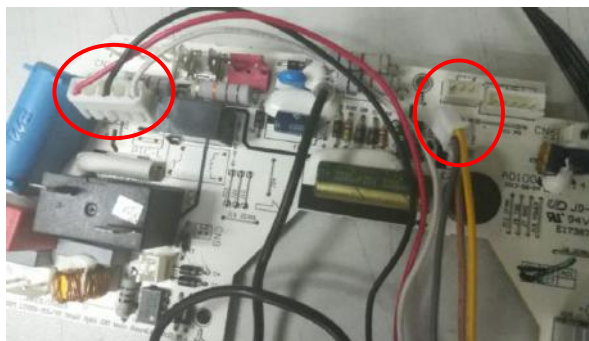
- 3) The PCB fail.

Solution: Replace the indoor main PCB.

10.5.2 Display E6

Reasons:

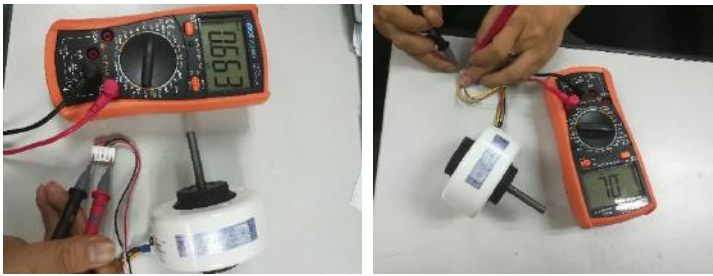
- 1) The indoor motor connection terminal loose or not plugged in.



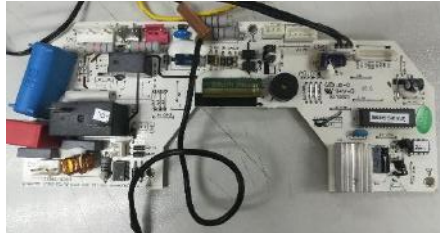
Solution: Check the connector of indoor motor and slot (CN3) and (CN4), if loose or not plugged in, please connect again.

- 2) The indoor motor damage.

Solution: Check and replace the motor.



3) The indoor main PCB damage.



Solution: Replace the indoor main PCB.

10.5.3 Display E3, E7

Reasons:

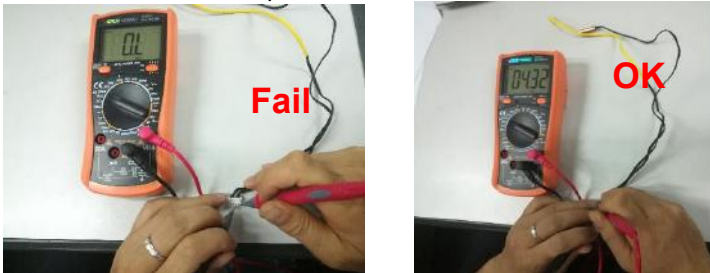
- 1) Outdoor pipe temp sensor, outdoor temp sensor & exhaust temp sensor connection terminal loose or not plugged in.



Solution: Check the connector of sensor and slot, if loose or not plugged in, please connect again.

- 2) Outdoor pipe temp sensor, outdoor temp sensor & exhaust temp sensor damage (short or broken).

Solution: Check and replace the sensor.



3) Outdoor PCB damage.

Solution: Check and replace the outdoor PCB.

10.5.4 Display E8

Reasons:

- 1) Outdoor discharge pipe temp sensor connection terminal loose or not plugged in.



Solution: Check the connector of sensor and slot, if loose or not plugged in, please connect again.

2) Outdoor pipe temp sensor damage.



Solution: Check and replace the sensor.

3) Outdoor PCB damage.

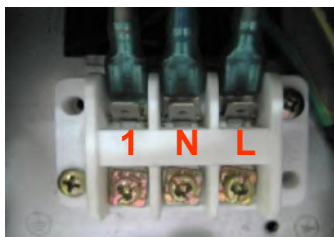
Solution: Check and replace the outdoor PCB.

10.5.5 Display E0, E5

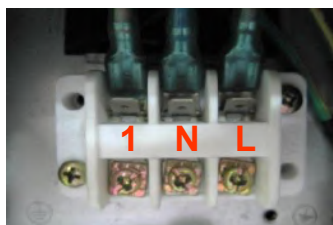
Reason: Indoor / outdoor communication damage.

Solution:

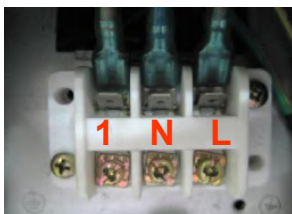
- 1) Check if the indoor and outdoor connections are correct. The terminal L and N which connect to indoor unit shall correspond to each other on indoor and outdoor units. Measure the voltage on outdoor terminal L and N (before display of E0 fault). If the voltage is "0", please replace indoor main PCB.



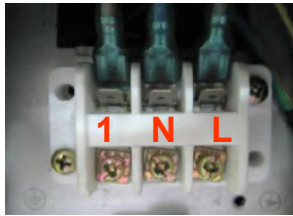
- 2) If the 1 & N which connect to indoor unit voltage is normal, measure the voltage between the outdoor terminal 1 and N. If the voltage change occurs between 0~24V (change pulse voltage), please replace outdoor PCB.



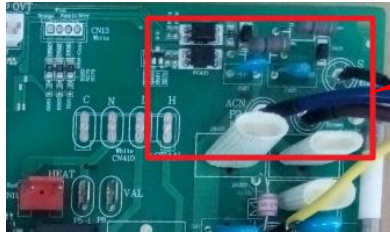
- 3) If the 1 & N which connect to indoor unit voltage is normal, measure the voltage between the outdoor terminal 1 and N. If the voltage change occurs between 0~12V (change pulse voltage), but there is no 24V, please replace the indoor PCB.



- 4) If the 1 & N voltage is normal, measure the voltage between the outdoor terminal 1 and N. If the voltage has no change, firstly replace the indoor main PCB. If the fault remains unsolved, replace the outdoor PCB.



5) Communication fault if resistance or optical coupler broken, replace outdoor PCB.



Communication circuit

6) Is there any burnt on PCB? If no, test the rectifier, FRD, IGBT etc. any component broken, replace PCB.

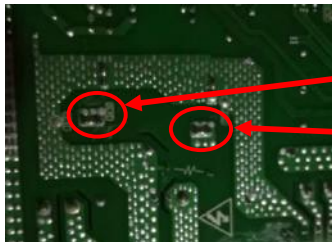


Any of 2 pins should be no short circuit

Any of 2 pins should be no short circuit

Any of 2 pins should be no short circuit

7) Test the DC voltage between DC+ and DC-. If the voltage is less than 50V approximately, please replace outdoor PCB.



DC-

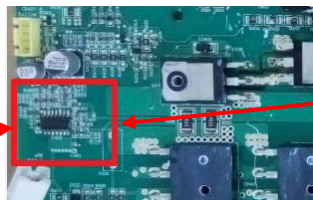
DC+

10.5.6 Display EA

Reason: Current sensor fault.

Solution:

- 1) Check for refrigerant leakage, to find the leakage point and recharge the refrigerant.
- 2) Current sampling circuit broken on the outdoor PCB and replace outdoor PCB.



Current sampling circuit

10.5.7 Display E9 (first display P0 or P9)

Reason: Outdoor PCB drive circuit damage.

Solution:

Re-energize and check the protection code on display. Firstly display P0.

- 1) If this code is displayed when the compressor is started for several seconds or even not started, check the compressor connection for correctness, if no insert wrong, replace outdoor PCB.



Compressor connection

- 2) Check if the outdoor module is tightly installed onto the radiating fins and if the silicone is applied evenly, fix the screws again if loose.



Heat radiation problem easily happened while the screw is not fixed tightly.

- 3) Check the system pressure, recharge refrigerant if the pressure is low, and discharge some refrigerant if the pressure is too high.
- 4) Check the outdoor ventilation and if there is any obstruction that affects the normal radiating of the air conditioner, and installation again.
- 5) If the above inspections are normal, but the fault remains unsolved, please replace the outdoor PCB.
Re-energize and check the protection code on display. Firstly display P9.
- a) Check the U,V, W connection, if is correctness or loose please connect again.



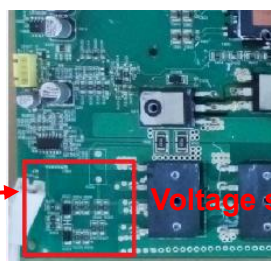
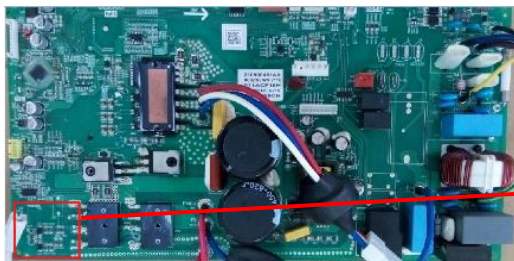
U, V, W connection

- b) If this code is displayed when the compressor is started for several seconds or even not started, check the compressor connection for correctness, if no insert wrong, replace outdoor PCB.

10.5.8 Display EU.

Reason: Voltage sensor damage.

Solution: Please replace outdoor PCB.

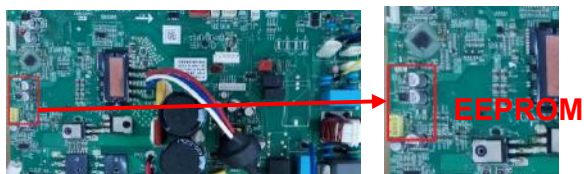


Voltage sensor

10.5.9 Display EE

Reason: EEPROM fault.

Solution: Shut down power supply and reenergize it, if the fault remains there, check the EEPROM installation, if no problem, please replace outdoor PCB.



10.5.10 Display CL

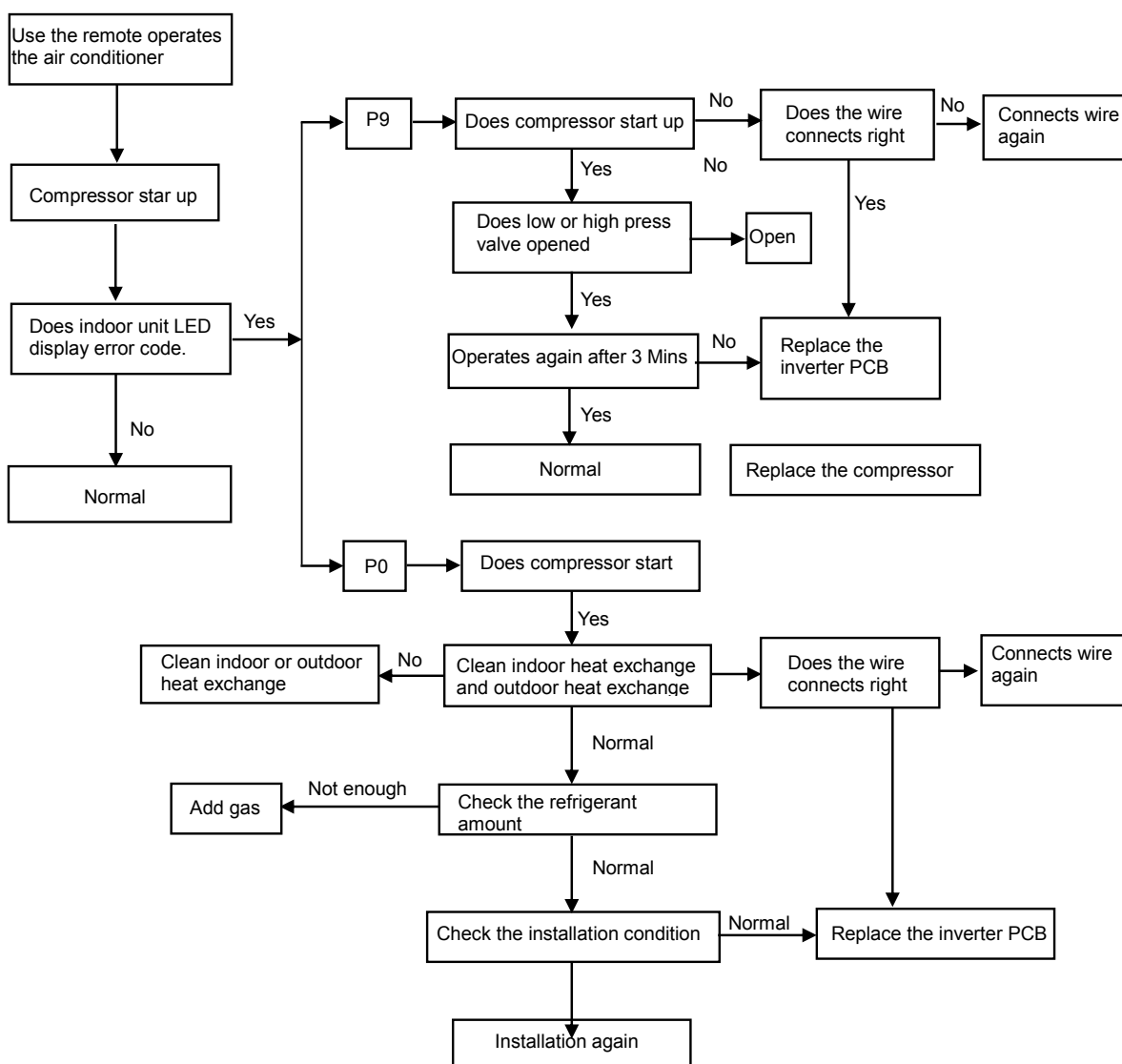
Cause: Air filter dirty, the unit operation Cumulative time 500 hours, the controller will confirmation the filter dirty and display CL.

Solution: Clean the air filter then cut off the power and supply again.

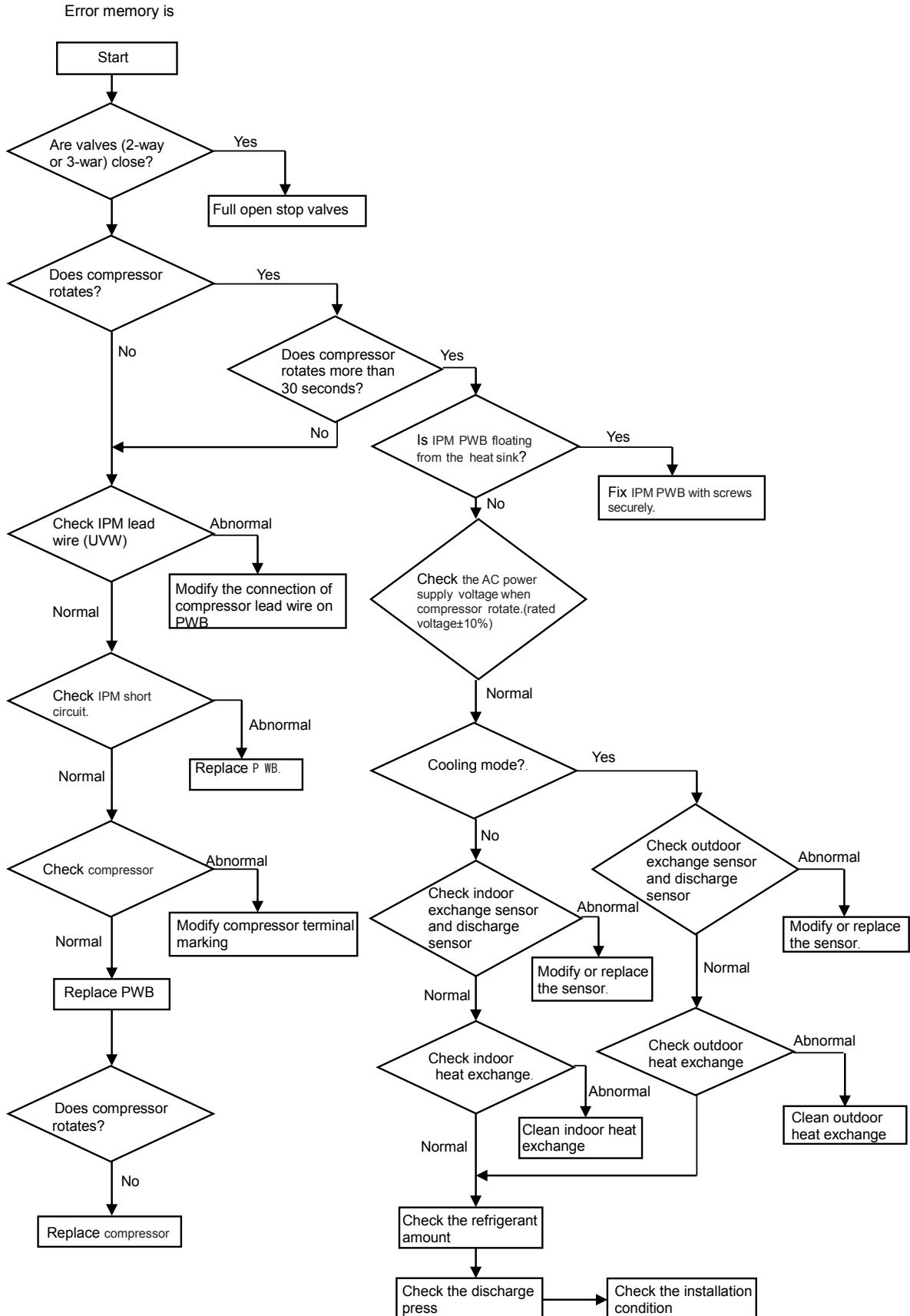
10.6 MALFUNCTION (PARTS) CHECK METHOD

10.6.1 Procedure for determining defective outdoor unit IPM/compressor

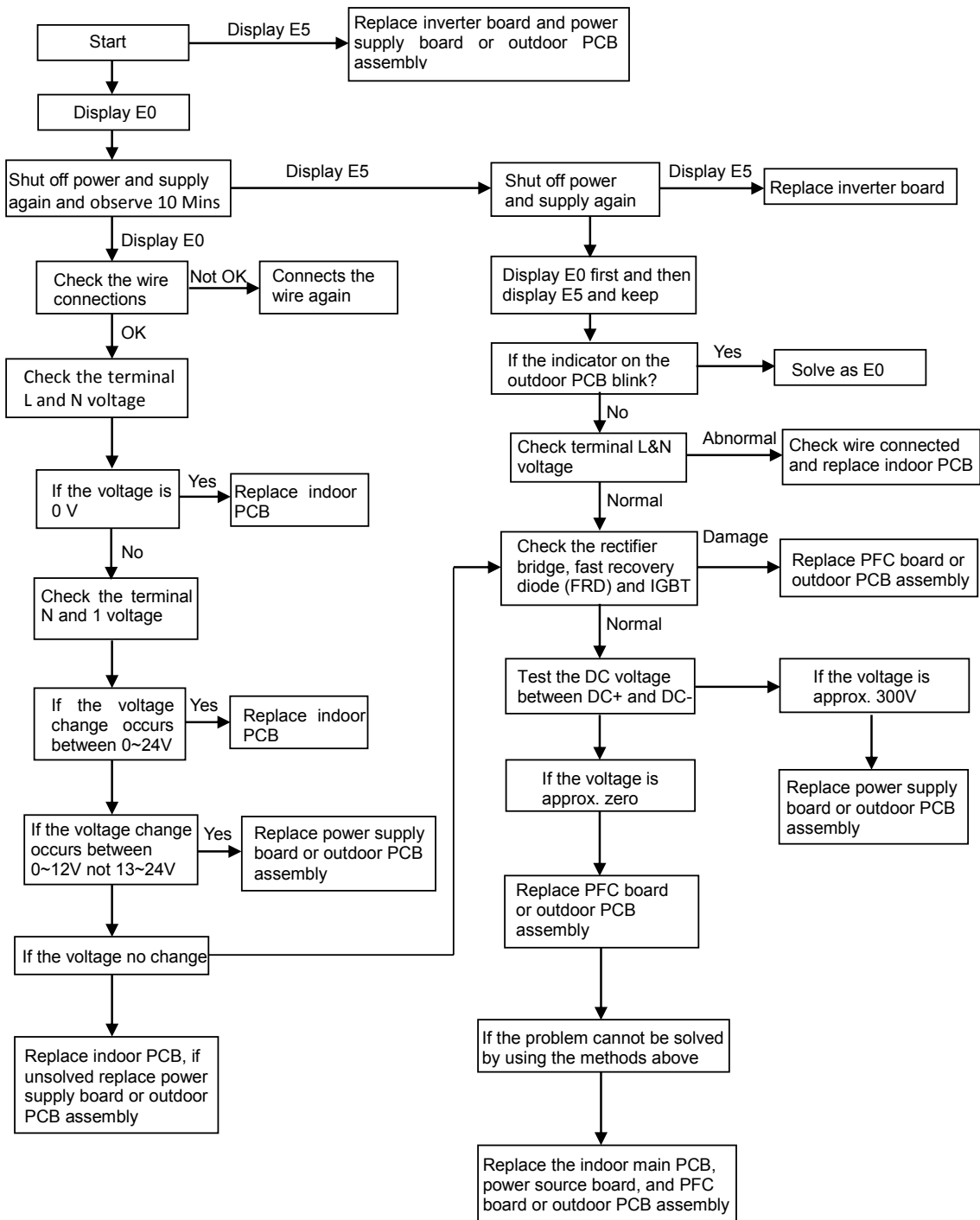
The following flow chart shows a procedure for locating the cause of a malfunction when the compressor does not start up and a DC overcurrent indication error occurs.



10.6.2 DC Over Current Error



10.6.3 E0,E5 Error

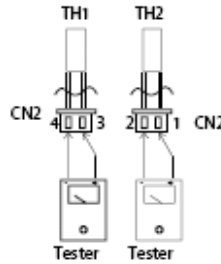


10.7 THERMISTOR TEMPERATURE CHARACTERISTICS

1. Indoor unit and outdoor exchange temperature and outside air temperature sensor temperature characteristics

TEMP. (°C)	Resistance (k Ohm)	Voltage of resistance	TEMP. (°C)	Resistance (k Ohm)	Voltage of resistance	TEMP. (°C)	Resistance (k Ohm)	Voltage of resistance
-30	63.513	4.628	15	7.447	2.968	60	1.464	1.115
-29	60.135	4.609	16	7.148	2.918	61	1.418	1.088
-28	56.956	4.589	17	6.863	2.868	62	1.374	1.061
-27	53.963	4.568	18	6.591	2.819	63	1.331	1.035
-26	51.144	4.547	19	6.332	2.769	64	1.290	1.009
-25	48.488	4.524	20	6.084	2.720	65	1.250	0.984
-24	45.985	4.501	21	5.847	2.671	66	1.212	0.960
-23	43.627	4.477	22	5.621	2.621	67	1.175	0.936
-22	41.403	4.452	23	5.404	2.572	68	1.139	0.913
-21	39.305	4.426	24	5.198	2.524	69	1.105	0.890
-20	37.326	4.399	25	5.000	2.475	70	1.072	0.868
-19	35.458	4.371	26	4.811	2.427	71	1.040	0.847
-18	33.695	4.343	27	4.630	2.379	72	1.009	0.825
-17	32.030	4.313	28	4.457	2.332	73	0.979	0.805
-16	30.458	4.283	29	4.292	2.285	74	0.950	0.785
-15	28.972	4.252	30	4.133	2.238	75	0.922	0.765
-14	27.567	4.219	31	3.981	2.192	76	0.895	0.746
-13	26.239	4.186	32	3.836	2.146	77	0.869	0.728
-12	24.984	4.152	33	3.697	2.101	78	0.843	0.710
-11	23.795	4.117	34	3.563	2.057	79	0.819	0.692
-10	22.671	4.082	35	3.435	2.012	80	0.795	0.675
-9	21.606	4.045	36	3.313	1.969	81	0.773	0.658
-8	20.598	4.008	37	3.195	1.926	82	0.751	0.641
-7	19.644	3.969	38	3.082	1.883	83	0.729	0.625
-6	18.732	3.930	39	2.974	1.842	84	0.709	0.610
-5	17.881	3.890	40	2.870	1.800	85	0.689	0.595
-4	17.068	3.850	41	2.770	1.760	86	0.669	0.580
-3	16.297	3.808	42	2.674	1.720	87	0.651	0.566
-2	15.565	3.766	43	2.583	1.681	88	0.633	0.552
-1	14.871	3.723	44	2.494	1.642	89	0.615	0.538
0	14.212	3.680	45	2.410	1.604	90	0.598	0.525
1	13.586	3.635	46	2.328	1.567	91	0.582	0.512
2	12.991	3.590	47	2.250	1.530	92	0.566	0.499
3	12.426	3.545	48	2.174	1.495	93	0.550	0.487
4	11.889	3.499	49	2.102	1.459	94	0.535	0.475
5	11.378	3.452	50	2.032	1.425	95	0.521	0.463
6	10.893	3.406	51	1.965	1.391	96	0.507	0.452
7	10.431	3.358	52	1.901	1.357	97	0.493	0.441
8	9.991	3.310	53	1.839	1.325	98	0.480	0.430
9	9.573	3.262	54	1.779	1.293	99	0.467	0.419
10	9.174	3.214	55	1.721	1.262	100	0.455	0.409
11	8.795	3.165	56	1.666	1.231			
12	8.433	3.116	57	1.613	1.201			
13	8.089	3.067	58	1.561	1.172			
14	7.760	3.017	59	1.512	1.143			

Resistance at 25°C : 5 kΩ.



TH1: indoor room temperature sensor and outside air temperature sensor

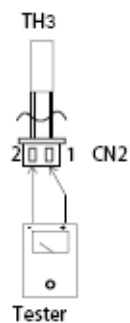
TH2: indoor exchange temperature sensor and outside exchange temperature sensor

Before measuring resistance, disconnect connectors as shown above.

2. Outdoor unit sensor temperature characteristics

TEMP. (°C)	R min (k Ohm)	R(t) (k Ohm)	R max (k Ohm)	TEMP. (°C)	R min (k Ohm)	R(t) (k Ohm)	R max (k Ohm)	TEMP. (°C)	R min (k Ohm)	R(t) (k Ohm)	R max (k Ohm)
-30	283.3	322.9	367.7	24	19.36	20.89	22.52	78	2.563	2.654	2.745
-29	267.4	304.4	346.3	25	18.55	20	21.54	79	2.481	2.567	2.654
-28	252.5	287.1	307.4	26	17.77	19.14	20.6	80	2.402	2.484	2.567
-27	238.5	270.9	307.4	27	17.03	18.32	19.7	81	2.327	2.404	2.483
-26	225.4	255.7	289.8	28	16.32	17.55	18.85	82	2.254	2.327	2.401
-25	213.1	241.4	273.3	29	15.65	16.81	18.04	83	2.183	2.253	2.323
-24	201.5	228	257.9	30	15	16.1	17.27	84	2.115	2.182	2.248
-23	190.6	215.5	243.4	31	14.39	15.43	16.54	85	2.05	2.113	2.176
-22	180.3	203.6	229.8	32	13.81	14.79	15.34	86	1.985	2.047	2.109
-21	170.7	192.5	217	33	13.25	14.18	15.17	87	1.922	1.983	2.045
-20	161.6	182.1	205	34	12.72	13.6	14.54	88	1.861	1.922	1.983
-19	153.1	172.3	193.7	35	12.21	13.05	13.93	89	1.802	1.862	1.923
-18	145	163.1	183.2	36	11.72	12.52	13.36	90	1.746	1.805	1.865
-17	137.5	154.4	173.2	37	11.26	12.01	12.81	91	1.692	1.75	1.809
-16	130.3	146.2	163.9	38	10.82	11.53	12.29	92	1.639	1.697	1.755
-15	123.6	138.5	155.1	39	10.29	11.07	11.78	93	1.589	1.646	1.703
-14	117.3	131.3	146.8	40	9.986	10.63	11.31	94	1.54	1.596	1.653
-13	111.3	124.4	139	41	9.6	10.21	10.85	95	1.493	1.549	1.604
-12	105.6	118	131.7	42	9.231	9.813	10.42	96	1.448	1.502	1.558
-11	100.3	111.9	124.7	43	8.878	9.43	10	97	1.404	1.458	1.512
-10	95.24	106.2	118.2	44	8.54	9.064	9.612	98	1.362	1.415	1.469
-9	90.49	100.8	112.1	45	8.217	8.714	9.233	99	1.321	1.373	1.426
-8	85.99	95.68	106.3	46	7.908	8.38	8.872	100	1.284	1.335	1.387
-7	81.75	90.86	100.8	47	7.612	8.06	8.526	101	1.245	1.296	1.348
-6	77.74	86.31	95.74	48	7.328	7.754	8.196	102	1.209	1.258	1.309
-5	73.94	82.01	90.88	49	7.057	7.461	7.88	103	1.173	1.222	1.272
-4	70.35	77.95	86.29	50	6.797	7.18	7.578	104	1.139	1.187	1.236
-3	66.96	74.11	81.96	51	6.548	6.912	7.289	105	1.105	1.153	1.202

-2	63.74	70.48	77.87	52	6.309	6.655	7.013	106	1.073	1.12	1.168
-1	60.69	67.05	74	53	6.08	6.409	6.748	107	1.042	1.089	1.136
0	57.81	63.8	70.34	54	5.861	6.173	6.495	108	1.013	1.058	1.104
1	55.08	60.72	66.88	55	5.651	5.947	6.253	109	0.9833	1.028	1.074
2	52.49	57.81	63.61	56	5.449	5.73	6.02	110	0.9553	0.9997	1.045
3	50.03	55.05	60.52	57	5.255	5.522	5.798	111	0.9283	0.9719	1.016
4	47.71	52.44	57.59	58	5.07	5.323	5.585	112	0.9021	0.9451	0.9892
5	45.5	49.97	54.82	59	4.891	5.132	5.381	113	0.8765	0.9191	0.9626
6	43.41	47.62	52.2	60	4.72	4.949	5.101	114	0.8524	0.894	0.9367
7	41.42	45.4	49.71	61	4.556	4.774	4.997	115	0.8087	0.8595	0.9117
8	39.53	43.2	42.33	62	4.398	4.605	4.817	116	0.8059	0.8461	0.8875
9	37.74	41.29	45.12	63	4.247	4.448	4.644	117	0.7837	0.8233	0.8641
10	36.04	39.39	43.01	64	4.101	4.288	4.479	118	0.7623	0.8012	0.8413
11	34.42	37.59	41	65	3.961	4.139	4.32	119	0.7415	0.7798	0.8193
12	32.89	35.87	39.1	66	3.827	3.995	4.167	120			
13	31.43	34.25	37.29	67	3.698	3.858	4.021	121	0.702	0.7386	0.7773
14	30.04	32.71	35.58	68				122	0.6631	0.7195	0.7572
15	29.72	31.24	33.95	69				123	0.6649	0.7007	0.7378
16				70	3.339	3.476	3.616	124	0.6472	0.6824	0.7189
17				71	3.229	3.359	3.491	125	0.6301	0.6647	0.7006
18	25.13	27.26	29.55	72	3.122	3.246	3.372	126	0.6135	0.6476	0.6829
19	24.05	26.07	28.23	73	3.02	3.138	3.257	127	0.5974	0.6309	0.6657
20	23.02	24.93	26.97	74	2.921	3.033	3.146	128	0.5818	0.6148	0.649
21	22.04	23.84	25.77	75	2.827	2.933	3.04	129	0.5667	0.5991	0.6328
22	21.1	22.81	24.63	76	2.735	2.836	2.938	130	0.5521	0.5839	0.6171
23	20.21	21.83	23.55	77	2.647	2.743	2.84				

R—Resistance**Resistance at 25°C : 20 kΩ****TH3: Outdoor unit discharge pipe sensor****Before measuring resistance, disconnect connectors as shown above.**